Sequential Spectral–Spatial Feature Convolution Network With Self-Attention for Remote Sensing Hyperspectral Image Classification

Jiqing Liu, Han Wang, *Member, IEEE*, Renhe Liu, *Student Member, IEEE*, Shaochu Wang, *Student Member, IEEE*, and Yu Liu, *Member, IEEE*

***Abstract*— The rich spatial and spectral information in hyperspectral images (HSIs) makes spectral–spatial relationships essential for HSI classification (HSIC). Recent advancements indicate convolutional neural networks (CNNs) excel in HSIC but often struggle with precise spectral feature extraction. Moreover, the abundance of spectral information presents challenges in efficient feature representation and minimizing cross-domain interference. To address these limitations, we propose an efficient sequential spectral–spatial feature convolution network (S^3FCN), employing successive subnetworks for spectral and spatial feature extraction with depthwise separable convolution. This approach balances the preservation of deep spectral and spatial features while significantly reducing network parameters, enhancing both performance and computational efficiency. We also introduce a sequential spectral–spatial attention module (S^3AM) to integrate cross-domain correlations. This module utilizes spectral features from the preceding subnetwork and multilevel residual layers for in-depth exploration of spatial features, enabling deep integration for improved classification performance. The proposed architecture's effectiveness is verified on five benchmark HSI datasets, including Pavia University, Salinas Valley, Kennedy Space Center, Indian Pines, and Houston 2013. Experimental results demonstrate that the sequential spectral–spatial connection in the feature extraction and attention mechanism integrated with depthwise separable convolution collectively surpasses current state-of-the-art (SOTA) techniques in classification accuracy with overall accuracies of 98.28%, 97.63%, 99.31%, 96.72%, and 95.38% across different datasets, while limiting the computation overhead, ensuring balanced network efficiency.**

***Index Terms*— Depthwise separable convolution, feature interaction, hyperspectral image classification (HSIC), self-attention, sequential spectral–spatial feature convolution.**

I. Introduction

HYPERSPECTRAL image classification (HSIC) is a crucial technique in remote sensing for land classification and object recognition. Unlike RGB remote sensing images with only three spectral bands, hyperspectral images (HSIs) capture hundreds or thousands of narrow, successive bands, providing rich spectral information. This detailed spectral data makes HSIs highly effective in agriculture [1], geological exploration [2], environmental monitoring [3], and soil organic carbon prediction [4].

In early research, machine learning (ML) methods were primarily used for feature extraction and classifier training, relying on expert knowledge for designing hand-crafted filters and optimization algorithms [5]. To address spectral and spatial redundancy, dimensionality reduction (DR) techniques were applied to project data into a lower-dimensional subspace [6], [7], [8], [9]. Despite the effectiveness of ML methods in HSI representation, hand-crafted features are limited by strong spectral correlation [10] and the curse of dimensionality, which compromises linear methods' assumptions in high-dimensional feature spaces [11]. Deep learning (DL) methods, such as CNNs, offer significant advantages by automatically learning high-level linear and nonlinear features from raw HSI data [12], [13], [14]. Many studies investigate deep CNNs to enhance DR in spectral feature processing, using high-performance ML techniques as preprocessing [15] and applying 1-D or combined 1-D and 2-D convolutional kernels to capture spectral variations [16], [17]. HSIs exhibit strong spatial–spectral correlation, where spatial features capture contextual relationships, enhancing the identification of spectral signatures. DR methods like principal component analysis (PCA) are often applied before spatial analysis to streamline feature extraction [18], [19], [20]. Advanced spatial analyses, including sparse representation [21], and deep feature extraction with CNNs [22], [23], [24] and transformers [25], [26], [27], further enrich spatial-domain processing.

Considering the strong correlation between spectral and spatial domains in HSIs, it is natural to explore the deep representation of spectral–spatial combined features to enhance the performance of HSIC. Such joint feature representation can be categorized into three types of architectures [28]: spectral preprocessing networks, spectral–spatial fusion networks, and spectral–spatial integrated networks. The spectral preprocessing structures consider flattening the spatial neighborhood to create stacked [29], [30], [31] or averaged [32], [33] spectral vectors. The spatial-learning networks are usually consecutively connected to exploit the spatial information

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from the preprocessed spectral features [34], [35], [36]. The spectral–spatial fusion networks involve separate branches to obtain deep spectral and spatial features to be further fused [37]. Usually, each branch is adapted to domain characteristics by employing the 1-D CNN for the spectrum and the 2-D CNN for the space [38], [39], [40], while recent works also present identical structures for both paths [41], [42]. Recent studies also propose transformer-based dual-branch designs by domain-related token expression [43] or attention [44]. The spectral–spatial integrated networks consider the direct joint representation of features coupled cross-domain. Initially, 2-D convolution was applied to only spatially process the HSI data cube, with spectral features being roughly compressed [22]. Later, 3-D convolution was involved in offering simultaneous spectral–spatial processing [45], [46]. Further developments promote variations like 3-D–2-D hybrid methods [47]. Recent studies explore transformers for cross-domain feature integration, using the spectral query to compress spatial redundancy [48], and the combination of CNNs and transformers for the synchronized improvement of local–global feature expression [49]. However, 3-D convolutions or transformers unavoidably raise the computation overhead, and to counter the issue, lightweight techniques [50], [51], [52] are designed to globally decompose the spectral and spatial domains and separately perform depthwise or pointwise convolution.

To sum up, extensive research has validated the exceptional performance of both CNN and transformer architectures in HSIC tasks, each with distinct advantages. CNNs are known for their simplicity and ease of use, making them suitable for convenient deployment, while transformers excel in capturing long-range dependencies and modeling complex relationships. In particular, CNN's efficiency remains advantageous for spectral–spatial cross-domain analysis. The literature review reveals that deep spectral feature extraction plays a crucial role in HSI analysis, as it directly influences the accuracy of subsequent spatial segmentation. Moreover, effective feature representation requires comprehensive strategies to allow cross-domain correlation and reduction of the network's complexity. To address these challenges, this article proposes a sequential CNN structure that effectively exploits deep HSI features with cross-domain attention. The main contributions of this article are summarized as follows.

1) We propose a novel and efficient sequential spectral–spatial feature convolution network (S^3FCN) that extracts cross-domain features while considering their correlation. This cascaded network follows a spectral-first, spatial-later feature extraction approach that captures and integrates cross-domain information to guarantee high classification accuracy. We also introduce 1-D and 2-D depthwise separable convolutions to reduce the number of parameters to improve efficiency.
2) We present a dedicated spectral feature extraction subnetwork for detailed compression of spectral redundancy. Utilizing a residual multilayer structure, it effectively extracts spectral features while preserving correlated spatial attributes, enabling the generation of pixel-level high-quality HSI labels for subsequent spatial-domain processing.
3) We develop a sequential spectral–spatial attention module (S^3AM) to establish a cross-domain attention mechanism. This module preserves spectral attention from the prior spectral subnetwork while extracting spatial features in the following, enabling layer-wise deep integration of cross-domain features.
4) The proposed method demonstrates superior generalization ability through comprehensive tests on five benchmark HSI datasets, spanning urban, rural, and urban–rural landscapes. In addition, comparisons with 13 advanced methods show that the proposed strategy offers a satisfactory balance between performance and efficiency.

The rest of this article is organized as follows. Section II discusses the related work. Section III describes the proposed classification network, highlighting the feature processing strategy. Section IV describes the experimental environment and state-of-the-art (SOTA) methods for comparison. Section V presents all results on benchmark datasets and gives detailed analyses. Section VI concludes this article.

II. Related Work

A. Advances in Spectral–Spatial Feature Extraction Methods

As reviewed in Section I, IID or 3-D CNNs can effectively produce joint representations of spectral–spatial features. Lee and Kwon [22] introduced a contextual deep CNN (CDCNN) that expands the spectral–spatial contextual feature extraction with multiscale filter banks. To enhance cross-domain deep feature extraction, Zhong et al. [45] suggested a spectral–spatial residual network (SSRN) that uses 3-D convolution kernels for global compression in the spectral domain and detailed computation of spatial features. However, 3-D convolution significantly increases the network's computation load, and later Roy et al. [47] proposed a hybrid spectral CNN (HybridSN) that partially uses consecutive multiple 3-D convolutions to get cross-domain joint features, along with a 2-D convolution layer to further extract the spatial characteristics, with multiple fully connected (FC) layers to assist classification. Recent works further emphasize spectral preprocessing. Li et al. [53] proposed an adaptive mask sampling and manifold to Euclidean subspace learning (AMS-M2ESL) network, suggesting a distance-based method to create a dual-channel spectral distance covariance representation to model linear and nonlinear interdependence in the spectral domain. Recently, Lu et al. [40] analyzed the Pearson correlation coefficients among adjacent spectral bands to design a grouped multiattention network (GMA-Net). All spectral bands are divided into multiple groups by spectral similarities, and groupwise spectral–spatial features are separately computed and fused to improve classification.

B. Application of Attention Mechanisms

Extensive methods are being considered when designing attention mechanisms to improve intra- and interdomain feature extraction. These methods are incorporated into backbones built with either CNNs or transformers. Roy et al. [54] proposed an attention-based adaptive spectral–spatial

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kernel ResNet (A^2S^2K-ResNet) that uses split 3-D convolution kernels for selective spectral and spatial feature fusion and recalibrated spectral attention to emphasize the interdependency between spectral bands. Recently, Fu et al. [55] designed a local–global gated CNN (LGG-CNN), introducing a novel attention computation as the local–global gating mechanism in the pathway for deeper convolution. It defines local gating as 2-D convolution with self-attention and global gating as FC layer-activated weighting from the next layer, combining both for final classification.

Recently, the transformer structure has been adopted into HSI encoders for its advantages in handling long-range dependencies with multithreaded self-attention. Hong et al. [25] developed SpectralFormer to first demonstrate the effectiveness of the transformer in enhancing accuracy. Building on this, Shi et al. [26] proposed a multiscale hierarchical convolution-aided Fourierformer (MHCFormer) to improve local feature extraction. Chen et al. [48] proposed a spectral query spatial transformer (SQSFormer), which utilizes spectral information from the center pixel to query relevant spatial features from neighboring pixels, thereby preserving essential spectral features and reducing irrelevant information through center-focused attention. Zhao et al. [52] enhanced the traditional vision transformer (ViT) and introduced a groupwise separable convolutional ViT (GSC-ViT), reducing parameters while boosting efficiency and performance. Meng et al. [27] also addressed redundancy in ViTs by proposing a multiscale Super-token transformer (MS^2T^2) to combine convolution and multihead attention at multiple spatial granularities defined by localized pooling. Although effective, the multihead long-range self-attention mechanism in transformers inherently introduces high complexity. In contrast, CNNs offer a simpler and more efficient means of exploring cross-domain feature correlations and still hold considerable promise.

C. Network Efficiency

Current research in HSIC increasingly focuses on balancing model performance in terms of classification accuracy and computational efficiency. To enhance the multilayer propagation, many methods adapt the basic structure of ResNet [56] and develop various residual connections. Paoletti et al. [57] proposed a pyramidal residual network (PyResNet) that integrates interlayer residual connections along with pyramidal convolution kernels to expand the receptive field in a deeper depth. The 3-D network, SSRN [45], uses over ten convolution layers, and multiple long-range residual connections are applied to every other layer to facilitate gradient propagation. Some networks incorporate dense blocks [58] to enlarge the network depth while avoiding model saturation in training [59], [60].

III. Proposed Method

This section details the proposed classification method. First, the overall design strategy of the land-cover classification network is presented, followed by the design of the basic feature computation unit, spectral and spatial feature extraction subnetworks, and a feature attention module.

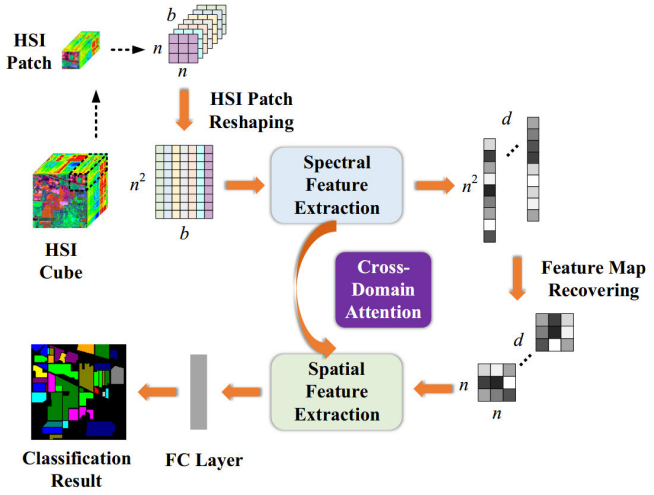


Fig. 1. Sketch diagram of the proposed network with sequentially connected subnetworks for spectral–spatial feature extraction through multilayer convolution. The cross-domain joint feature representation is facilitated by domain-linking attention.

A. Overall Network Architecture

An illustrative sketch of the proposed S^3FCN is shown in Fig. 1. The mainframe integrates two subnetworks in sequence to extract spectral and spatial features consecutively. For remote sensing HSIs, the spectral and spatial dimensions contain varying information levels, with higher redundancy in the spectral dimension due to its low-rank characteristics. To address this, we propose a sequential spectral–spatial feature extraction approach that first applies incremental compression to reduce spectral redundancy, followed by spatial feature extraction with spectral–spatial joint attention to capturing complex geomorphic features. The classification network is trained end-to-end using cross-entropy loss.

Specifically, each original 3-D HSI patch, $\mathbf{I}_{\mathrm{p}} \in \mathbb{R}^{n\times n\times b}$, with the spatial dimension of n and spectral bands of b, is transformed into a spectral tensor through the reshaping operation, $\mathcal{R}$

$$\mathbf{C} = \mathcal{R}(\mathbf{I}_{\mathrm{p}}) \tag{1}$$

where $\mathbf{C} \in \mathbb{R}^{n^2\times b}$ represents reformed hyperspectral tensor. The reshaped spectral inputs are processed with a spectral feature extraction subnetwork described in detail in Section III-C. The condensed information generated from the spectral feature learning subnetwork is expressed as $\mathbf{f}^{\mathrm{spectral}} \in \mathbb{R}^{n^2\times 1\times d}$, where d is the feature channel count. A feature map recovering function, $\mathcal{R}^{-1}$, is defined as the reverse operation of the patch reshaping, to recover the deep extracted spectral feature vector into a 2-D feature map, $\mathbf{F}^{\mathrm{spectral}} \in \mathbb{R}^{n\times n\times d}$

$$\mathbf{F}_i^{\mathrm{spectral}} = \mathcal{R}^{-1}\left(\mathbf{f}_i^{\mathrm{spectral}}\right) \tag{2}$$

where $i = 1, 2, \ldots, d$, and $\mathbf{F}_i^{\mathrm{spectral}} \in \mathbb{R}^{n\times n}$. Subsequently, the spatial feature learning subnetwork described in Section III-D further processes the recovered 2-D spectral feature maps for spatial characterization and produces the classification result through the FC layer. More importantly, a cross-domain feature integration module computes the spectral–spatial joint

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self-attention to maintain a close association of the spectral characteristics during the spatial feature processing, which will be discussed in Section III-E.

B. 1-D and 2-D Depthwise Separable Convolution

Conventional convolution often relies on expanding the number of channels to enhance deep feature representation, which increases parameter counts and computational complexity. Inspired by depthwise separable convolution (DSC) [61], we introduce 1D- and 2D-DSC to reduce convolution costs, where 1D-DSC extracts spectral features from reshaped spectral vectors, and 2D-DSC processes spatial features from recovered feature maps. Fig. 2 illustrates an example of 1D-DSC for spectral feature processing, which involves depthwise convolution for per-channel extraction followed by pointwise convolution to aggregate features across channels. For feature maps at the input, $\mathbf{F}_{\text{in}}$, and output, $\mathbf{F}_{\text{out}}$, the DSC procedure can be expressed as

$$\mathbf{F}_{\text{in}} = \mathcal{P}(\mathcal{D}(\mathbf{F}_{\text{out}})) \tag{3}$$

where $\mathcal{D}$ and $\mathcal{P}$ represent depthwise and pointwise convolution, respectively.

Define the output feature map size as $H_{\text{out}} \times W_{\text{out}}$, with input and output channel counts C_{in} and C_{out}, respectively. For 2D-DSC, the depthwise convolution kernel size is $k \times k$, and the pointwise convolution is $1 \times 1 \times C_{\text{in}}$. The parameter count, P_{DSConv}, and multiply-accumulate operations (MACs), M_{DSConv}, of the entire DSC are estimated as

$$P_{\text{DSConv}} = \left(k^2 + C_{\text{out}}\right) C_{\text{in}} \tag{4}$$

$$M_{\text{DSConv}} = \left(k^2 + C_{\text{out}}\right) C_{\text{in}} H_{\text{out}} W_{\text{out}} \tag{5}$$

while in conventional convolution, the parameter count, P_{conv}, and MACs, M_{conv}, are

$$P_{\text{Conv}} = k^2 C_{\text{out}} C_{\text{in}} \tag{6}$$

$$M_{\text{Conv}} = k^2 C_{\text{out}} C_{\text{in}} H_{\text{out}} W_{\text{out}}. \tag{7}$$

Thus, DSC reduces parameters and MACs by a ratio of $(k^2 C_{\text{out}}/(k^2 + C_{\text{out}}))$, significantly reducing computational complexity.

C. Spectral Feature Extraction with Multilevel Residuals

A detailed illustration of the network architecture is shown in Fig. 3. The first subnetwork focuses on the deep extraction and condensing of features in the spectral domain. The features are successively extracted through a multilayer convolutional structure for a deep representation. To avoid cross-interference in the spatial domain, the spectral convolution is restrained with depthwise convolution kernels of $1 \times k_{\text{c}}$, where k_{c} is the length along the spectral dimension.

The reshaped HSI patch is first processed with a traditional 1-D convolution, with the kernel length $k_{\text{c}} = 9$, stride $s = 2$, and no padding for shallow representation to extract a series of initial features with the channel number of 40. Next, deep feature extraction is conducted with four blocks, each comprising 1D-DSC, batch normalization (BN), and rectified linear unit (ReLU). Residual connections are

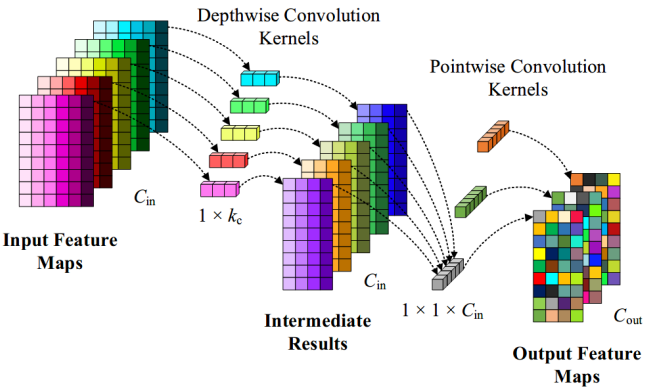


Fig. 2. Illustration of the proposed 1D-DSC with $1 \times k_{\text{c}}$ depthwise kernels and $1 \times 1 \times C_{\text{in}}$ pointwise kernels.

added at intervals between every two convolution layers after the ReLU activation, allowing for deep extraction stability and the representation of nonlinear interlayer relations. Each residual layer involves a DSC unit with the kernel length of $k_{\text{c}} = \{7, 9, 11, 13\}$ along the extraction depth. Padding is added to maintain the equal size of the input and output feature maps. Outputs from all internal residual layers contain an equal number of channels, $C_{\text{feature}} = 40$, which are then concatenated before final compression. A final DSC block serves as the feature condenser with the kernel length equal to the feature map's width to concentrate features into vectors of $\mathbf{f}^{\text{spectral}} \in \mathbb{R}^{n^2 \times 1 \times 200}$, which are then converted into a series of 2-D maps for spatial feature computation in the following.

To summarize, the residual convolution is expressed as

$$\mathbf{F}_i = \text{DSConv}(\mathbf{F}_{i-1}) + \mathbf{F}_{i-1} \tag{8}$$

where $i = 1, 2, 3, 4$. $\mathbf{F}_0$ is the output from the first conventional convolution block and DSConv$(\cdot)$ represents the consecutive operation of 1D-DSC, BN, and ReLU. Feature maps at each layer are all with the same dimension of $\mathbf{F}_{0,1,\ldots,4} \in \mathbb{R}^{n^2 \times m}$, where $m = \lfloor((b-9)/2)\rfloor + 1$, and b denotes the original spectral bands. The last block for feature compression is computed as

$$\mathbf{f}^{\text{spectral}} = \text{DSConv}(\text{Concat}(\mathbf{F}_0, \mathbf{F}_1, \ldots, \mathbf{F}_4)). \tag{9}$$

D. Spatial Feature Extraction With Cross-Domain Attention

The compressed and recovered spectral features are fed into the next sequentially connected subnetwork for deep spatial feature extraction. As shown in Fig. 3, this process involves three consecutive 2D-DSC blocks. Attention mechanisms are applied as interlayer and end-to-end modules to reinforce the previous layers' weighting. The attention module, S^{3}AM, computes global self-attention both spectral-wise and spatial-wise, allowing for a cooperative strengthening of cross-domain attributes, with the details discussed in Section III-E. To be specific, the input feature is first reshaped into the form of $n \times n \times 200$. Three consecutive DSC layers all incorporate 2-D depthwise kernels with the size of 3×3 for spatial filtering, and the feature channel dimension is controlled as $C_{\text{feature}} = \{60, 90, 120\}$. The convolution stride $s = 1$, and

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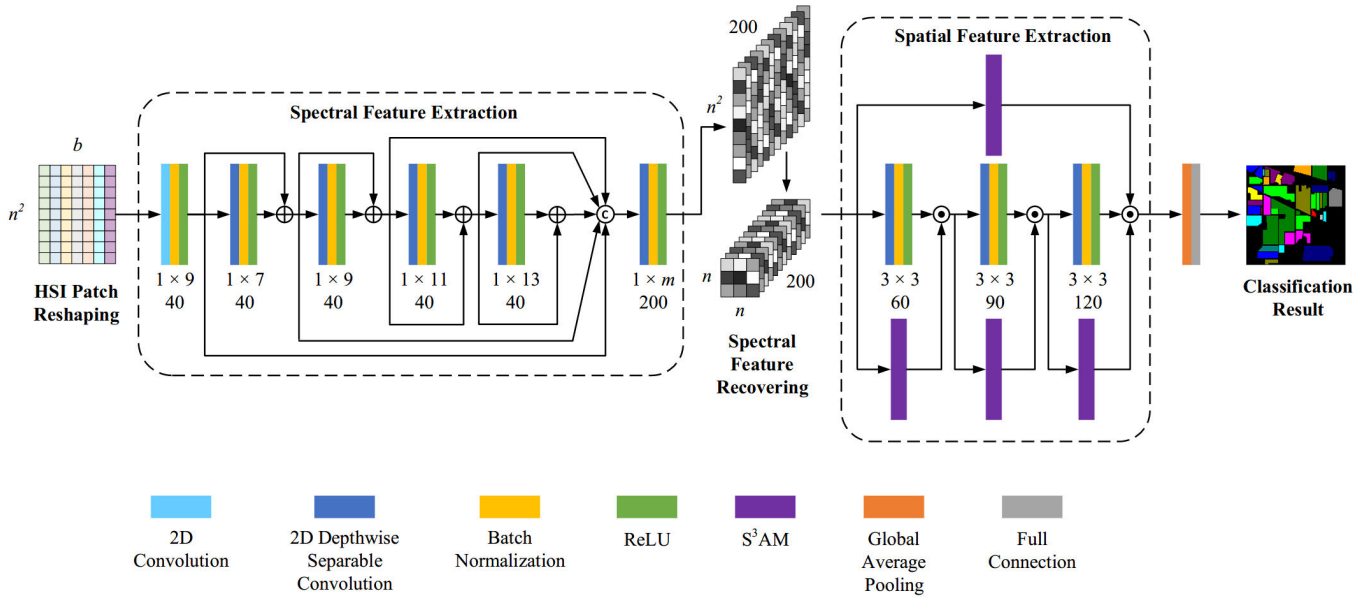


Fig. 3. Overall architecture of the proposed S^{3}FCN for HSIC. The network consists of a sequential connection of spectral and spatial feature extraction subnetworks for deep multidomain feature representation. Depthwise separable convolution is adopted as the basic convolution block.

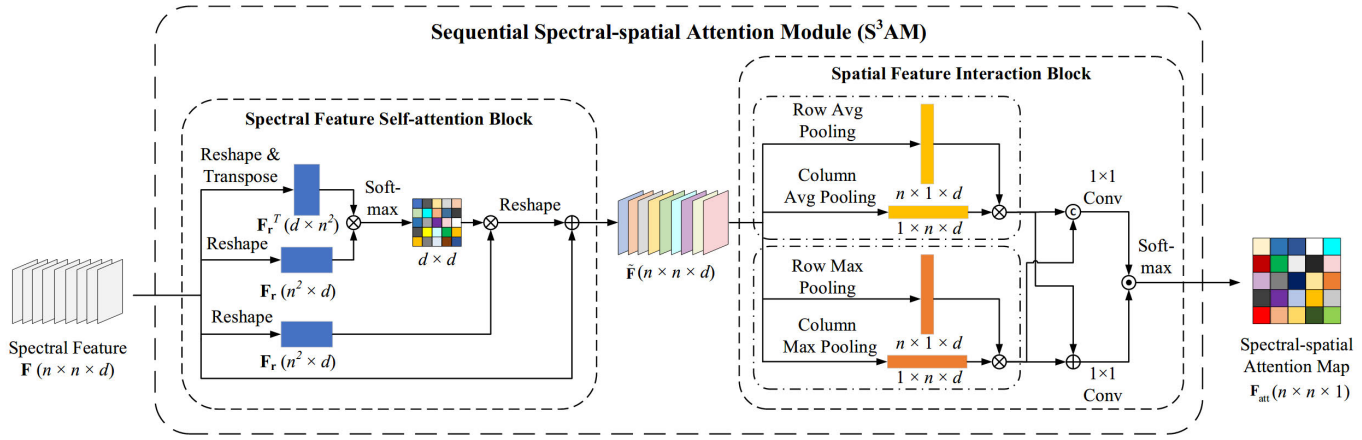


Fig. 4. Architecture of the proposed S^{3}AM for computation of spectral–spatial cross-domain attention in a sequential manner.

padding is applied to maintain the same feature map dimension in and out. At the output, a global average pooling converts the deep extracted spatial features into a condensed vector with the size of $1 \times 1 \times 120$, which is then processed with an FC layer to generate the final classification output.

E. Sequential Spectral–Spatial Attention Module

The attention module, S^{3}AM, is designed to produce integrated spectral–spatial attention maps. The aim is to integrate spectral channel relations into the spatial feature extraction process, avoiding the degradation of spectral information during spatial pixel weighting. Similar to the framework of the main network, this attention module also consists of two consecutive parts, as illustrated in Fig. 4. In general, multichannel spectral features are first processed by a spectral self-attention block to preserve spectral channel correlation without affecting spatial pixel continuity. This output is then fed into a spatial feature interaction block to examine the global interaction in the spatial domain. For comprehensiveness and computational efficiency, the inputs to the spatial attention block are processed with average and max-pooling in rows and columns, and the spatial interaction is evaluated with vector multiplication in either branch. The outputs from these two pooling branches are combined to obtain the final cross-domain attention map.

In detail, the feature map generated from the spectral feature extraction subnetwork, $\mathbf{F} \in \mathbb{R}^{n\times n\times d}$, is first reshaped into a 2-D tensor, $\mathbf{F}_\mathrm{r} \in \mathbb{R}^{n^2\times d}$. Then, channel-wise spectral self-attention is directly computed by applying interchannel weighting to encode the relations between adjacent spectral channels, which can be expressed as

$$\mathbf{F}_\mathrm{r} = \mathcal{R}(\mathbf{F}) \tag{10}$$

$$\tilde{\mathbf{F}} = \mathcal{R}^{-1}\left(\mathrm{Softmax}\left(\mathbf{F}_\mathrm{r}\mathbf{F}_\mathrm{r}^T/\sqrt{d}\right)\mathbf{F}_\mathrm{r}\right) + \mathbf{F}. \tag{11}$$

In the second step, the spatial feature self-attention is computed solely in the spatial dimension while keeping the channel dimension unchanged. The core idea is to encode the spatial

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TABLE I
DESCRIPTION OF BENCHMARK DATASETS: PU, SV, KSC, IP, AND HT

Dataset	Image Size (pixel)	Spatial Resolution	Spectral Range	Selected HSI Bands	Landform Categories
PU	610 × 340	1.3 m	430–860 μm	103	9
SV	512 × 217	3.7 m	400–2500 μm	204	16
KSC	512 × 614	18 m	400–2500 μm	176	13
IP	145 × 145	20 m	400–2500 μm	200	16
HT	349 × 1905	2.5 m	380–1050 μm	114	15

pixel global interaction across the row and column dimensions. Additionally, a dual-pooling mechanism employing both average and max-pooling is applied. In detail, the spectral attention map, $\tilde{\mathbf{F}} \in \mathbb{R}^{n\times n\times d}$, is converted into vectors via row and column pooling at each channel. Then, both paths involve row and column matrix products to generate spatial self-attention. The results from two pooling branches are combined through concatenation and element summation and processed with 1×1 convolution to compress all channels, giving the final spectral–spatial attention map, $\mathbf{F}_{\text{att}}$. We selected Softmax as the activation function through a comparison (Table S1) in the Supplementary Material. This computation process can be expressed as

$$\mathbf{F}_{\text{avg_att}} = \text{AvgPool}_{\text{row}}(\tilde{\mathbf{F}}) \times \text{AvgPool}_{\text{column}}(\tilde{\mathbf{F}}) \tag{12}$$

$$\mathbf{F}_{\text{max_att}} = \text{MaxPool}_{\text{row}}(\tilde{\mathbf{F}}) \times \text{MaxPool}_{\text{column}}(\tilde{\mathbf{F}}) \tag{13}$$

$$\mathbf{F}_{\text{att}} = \text{Softmax}\big(\text{Conv}_{1\times1}\big(\mathbf{F}_{\text{avg_att}} \oplus \mathbf{F}_{\text{max_att}}\big) \odot \text{Conv}_{1\times1}\big(\text{Concat}\big(\mathbf{F}_{\text{avg_att}}, \mathbf{F}_{\text{max_att}}\big)\big)\big). \tag{14}$$

IV. EXPERIMENTAL SETUP

This section outlines the experimental setup for validating the proposed network, including descriptions of five benchmark datasets, 13 SOTA comparison methods for performance evaluation, and the computation environment.

A. Benchmark Datasets

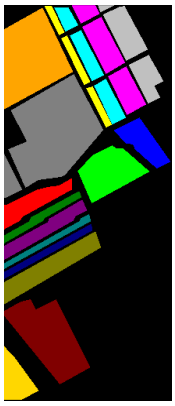
Five benchmark public datasets, including Pavia University (PU), Salinas Valley (SV), Kennedy Space Center (KSC), Indian Pines (IP), and Houston 2013 (HT) are selected for evaluation. Characteristics of the datasets are summarized in Table I and illustrated in Figs. 5–9. The PU and HT data are collected in the visible to near-infrared range, and the SV, KSC, and IP datasets cover the visible to short-wave infrared spectra. All datasets were preprocessed following the convention to remove noisy or water absorption bands. Regarding landform classes, PU is categorized as urban with many regular geometrical shapes. SV, IP, and KSC contain mostly rural objects, while the KSC data contains mostly small irregular blocks. The HT dataset involves urban and rural landscapes with scattered land objects. Notably, the IP and HT data are more challenging in HSIC because both datasets exhibit significant class imbalance in their land-cover pixel distributions.



Class	Color	Land cover	Samples
C1		Asphalt	6631
C2		Meadows	18649
C3		Gravel	2099
C4		Trees	3064
C5		Painted metal sheets	1345
C6		Bare Soil	5029
C7		Bitumen	1330
C8		Self-Blocking Bricks	3682
C9		Shadows	947
		TOTAL	42776

(a) (b)

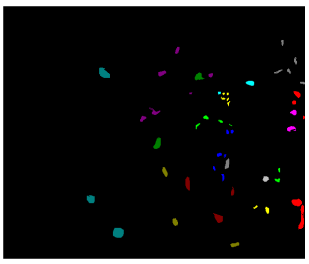
Fig. 5. Illustration of the PU dataset. (a) Ground truth. (b) Dataset labels.



Class	Color	Land cover	Samples
C1		Brocoil-green-weeds_1	2009
C2		Brocoil-green-weeds_2	3726
C3		Fallow	1976
C4		Fallow-rough-plow	1394
C5		Fallow-smooth	2678
C6		Stubble	3959
C7		Celery	3579
C8		Grapes-untrained	11271
C9		Soil-vinyard-develop	6203
C10		Corn-senesced-green-weeds	3278
C11		Lettuce-romaine-4wk	1068
C12		Lettuce-romaine-5wk	1927
C13		Lettuce-romaine-6wk	916
C14		Lettuce-romaine-7wk	1070
C15		Vinyard-untrained	7268
C16		Vinyard-vertical-trellis	1807
		TOTAL	54129

(a) (b)

Fig. 6. Illustration of the SV dataset. (a) Ground truth. (b) Dataset labels.



Class	Color	Land cover	Samples
C1		Scrub	761
C2		Willow-swamp	243
C3		CP-hammock	256
C4		Slash-pine	252
C5		Oak/Broadleaf	161
C6		Hardwood	229
C7		Swap	105
C8		Graminoid-marsh	431
C9		Spartina-marsh	520
C10		Cattail-marsh	404
C11		Salt-marsh	419
C12		Mud-flats	503
C13		Water	927
		TOTAL	5211

(a) (b)

Fig. 7. Illustration of the KSC dataset. (a) Ground truth. (b) Dataset labels.



Class	Color	Land cover	Samples
C1		Alfalfa	46
C2		Corn-notill	1428
C3		Corn-mintill	830
C4		Corn	237
C5		Grass/pasture	483
C6		Grass/trees	730
C7		Grass/pasture-mowed	28
C8		Hay-windrowed	478
C9		Oats	20
C10		Soybean-notill	972
C11		Soybean-mintill	2455
C12		Soybean-clean	593
C13		Wheat	205
C14		Woods	1265
C15		Bldg-Grass-Tree-Drivers	386
C16		Stone-Steel-Towers	93
		TOTAL	10249

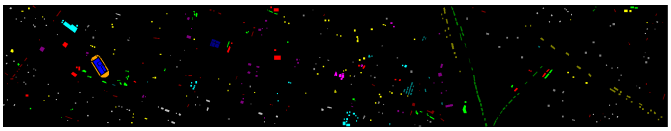
(a) (b)

Fig. 8. Illustration of the IP dataset. (a) Ground truth. (b) Dataset labels.

B. Comparative Methods

In total, 13 SOTA methods are compared to verify the effectiveness of the sequential spectral–spatial feature extraction

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(a)

Class	Color	Land cover	Samples	Class	Color	Land cover	Samples
C1		Healthy grass	1251	C9		Road	1252
C2		Stressed grass	1254	C10		Highway	1227
C3		Synthetic grass	697	C11		Railway	1235
C4		Trees	1244	C12		Parking Lot 1	1233
C5		Soil	1242	C13		Parking Lot 2	469
C6		Water	325	C14		Tennis Court	428
C7		Residential	1268	C15		Running Track	660
C8		Commercial	1244				
		TOTAL					15029

(b)

Fig. 9. Illustration of the HT dataset. (a) Ground truth. (b) Dataset labels.

and attention design of the proposed S^{3}FCN. Based on the major characteristics, the comparative methods can be roughly grouped into three categories.

1) *Integrated 3-D Convolution:* A^{2}S^{2}K-ResNet [54], HybridSN [47], SSRN [45].
2) *Separate Spectral–Spatial Feature Extraction and Fusion:* AMS-M2ESL [53], CDCNN [22], LGG-CNN [55], GMA-Net [40], and PyResNet [57].
3) *Transformer Methods:* GSC-ViT [52], MHC-Former [26], MS^{2}T^2 [27], SpectralFormer [25], and SQSFormer [48].

The classification accuracy is evaluated per class and on an overall scale. The overall metrics include overall accuracy (OA), average accuracy (AA), and Kappa coefficient (κ).

C. Computation Environment

The network is implemented in the PyTorch environment, with a training epoch of 200 and batch size 16. The Adam algorithm is used for optimization. During the experiment, the learning rate was tested within the range of {0.01, 0.005, 0.003, 0.001, 0.0005, 0.0003, 0.0001, 0.00005}. Detailed analyses for selecting the optimum learning rate and training epoch can be found in Tables S2 and S3 in the Supplementary Material. The optimum learning rate of 0.0003 was selected for HT and 0.0001 for the other four datasets. Each test result is an average of 30 experiments implemented with the same configuration. The hardware environment is established on a cost-effective laptop platform and includes an Intel Core i7-12700H CPU, an NVIDIA GeForce RTX 3060 6 GB Laptop GPU, and 32 GB RAM. In training, all input patches have the same 2-D size of 9×9 for a fair comparison. Training samples are randomly selected for each labeled category in the original dataset following a predefined training set ratio as the percentage of training samples to the total number of samples. Such ratios are determined through repeated pretests, with optimal values of 0.5%, 0.5%, 5%, 3%, and 2% for the PU, SV, KSC, IP, and HT datasets, respectively. The remaining samples are used as the test set. A verification study for the training set ratio selection is given in Section V-C. The details of dataset preparation are further shown in Table II.

TABLE II
NUMBER OF SAMPLES IN THE TRAINING AND TEST SETS ACROSS ALL DATASETS IN THE EXPERIMENTS

Class	Training Set					Test Set				
	PU	SV	KSC	IP	HT	PU	SV	KSC	IP	HT
Total	210	263	256	307	293	42566	53866	4955	9942	14736
C1	33	10	38	3	25	6598	1999	723	43	1226
C2	93	18	12	42	25	18556	3708	231	1386	1229
C3	10	9	12	24	13	2089	1967	244	806	684
C4	15	6	12	7	24	3049	1388	240	230	1220
C5	6	13	8	14	24	1339	2665	153	469	1218
C6	25	19	11	21	6	5004	3940	218	709	319
C7	6	17	5	3	25	1324	3562	100	25	1243
C8	18	56	21	14	24	3664	11215	410	464	1220
C9	4	31	26	3	25	943	6172	494	17	1227
C10	N/A	16	20	29	24	N/A	3262	384	943	1203
C11	N/A	5	20	73	24	N/A	1063	399	2382	1211
C12	N/A	9	25	17	24	N/A	1918	478	576	1209
C13	N/A	4	46	6	9	N/A	912	881	199	460
C14	N/A	5	N/A	37	8	N/A	1065	N/A	1228	420
C15	N/A	36	N/A	11	13	N/A	7232	N/A	375	647
C16	N/A	9	N/A	3	N/A	N/A	1798	N/A	90	N/A

V. RESULTS AND ANALYSIS

This section summarizes and discusses the experimental results, involving comparisons with 13 methods based on the metrics per class and overall, as well as computation efficiency. Ablation studies are also performed to examine the contribution of the attention module and the influence of different network training strategies.

A. Classification Results and Analysis

Classification results are compared among the above-listed methods over five datasets. Quantitative results are listed in Tables III–VII, with the bold and underscored texts representing the best and the second-best results.

1) Results of the PU Dataset: Classification results are shown in Table III and Fig. 10. In Table III, the proposed network achieved the best or the second-best in most categories compared with other methods. The overall metrics of OA, AA, and κ exceed the second-best network by 1.97%, 2.36%, and 2.62%, respectively. Categories C3 (gravel) and C8 (self-blocking bricks) show lower classification accuracy in general due to their similar spectral–spatial features, while our method performs better. The results conform with the comparison in Fig. 10 that the proposed method efficiently distinguishes various geometric shapes in urban landscapes while reducing classification errors for large areas.

2) Results of the SV Dataset: Classification results are shown in Table IV and Fig. 11. Table IV shows that the proposed method performs well in all categories and leads in all three overall metrics of OA, AA, and κ by 1.65%, 0.84%, and 1.84% to the second place, respectively. Compared with the PU dataset, the SV dataset introduces more spectral bands and extends the spectrum into the short-wave infrared to enhance the spectral coding of landforms. Our method can achieve better results for class C8 (grapes-untrained), whose landforms are similar to those of the adjacent land class C15

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TABLE III
CLASSIFICATION RESULTS OF THE PU DATASET WITH THE TRAINING SET RATIO OF 0.5%

Class	A^2S^2K-ResNet [54]	AMS-M2ESL [53]	CDCNN [22]	GMA-Net [40]	GSC-ViT [52]	HybridSN [47]	LGG-CNN [55]	MHCFormer [26]	MS^2T^2 [27]	PyResNet [57]	SpectralFormer [25]	SQSFormer [48]	SSRN [45]	S^3FCN
C1	95.03±1.68	95.37±2.03	83.13±3.71	90.97±3.94	95.30±2.45	88.11±2.99	86.92±5.93	91.80±2.86	94.92±2.22	72.72±6.18	78.07±3.72	90.46±3.91	96.01±2.55	**98.31±1.24**
C2	98.44±0.90	95.50±1.39	94.40±2.41	97.61±1.12	97.81±1.48	92.47±1.73	96.76±1.39	98.09±1.04	98.49±0.79	81.80±6.05	86.79±2.07	96.05±1.51	98.26±0.99	**99.35±0.41**
C3	85.64±7.69	92.58±4.61	45.92±7.50	71.33±8.96	92.00±4.98	65.31±9.13	71.02±11.36	81.70±8.03	91.38±5.88	32.09±18.75	40.85±6.22	73.29±10.45	88.43±8.36	**95.72±3.96**
C4	**99.16±1.12**	96.59±2.52	95.46±3.81	97.11±1.56	97.45±2.28	96.25±3.45	99.11±0.68	99.06±0.75	96.56±2.17	89.47±23.36	94.28±4.09	96.75±1.79	99.02±1.58	98.80±0.77
C5	99.35±0.76	**99.98±0.06**	97.29±3.78	99.51±0.86	99.13±1.00	96.46±4.25	99.18±1.14	99.17±1.50	97.51±2.83	97.53±5.48	94.38±3.85	99.69±0.38	99.66±0.59	99.54±0.41
C6	97.13±1.74	92.79±4.23	84.11±6.39	94.38±3.57	97.50±1.76	79.98±5.58	96.30±2.67	97.40±2.30	98.11±1.18	71.31±16.92	73.24±7.29	93.15±3.80	95.92±3.51	**98.47±1.68**
C7	93.95±6.02	96.18±3.44	66.55±12.05	84.84±11.04	93.30±5.57	69.51±14.33	57.30±36.54	86.09±8.12	96.96±5.45	37.30±24.78	44.43±10.97	81.47±10.05	92.47±6.09	**98.52±2.01**
C8	89.46±3.20	89.27±3.92	67.77±5.10	82.82±4.12	89.98±4.31	77.68±4.98	81.42±6.11	82.57±4.57	88.20±5.40	62.05±13.48	65.51±4.88	80.86±6.43	86.61±5.56	**93.50±3.90**
C9	98.49±1.82	99.37±1.55	89.94±9.23	97.07±2.37	97.40±1.27	91.81±7.15	97.22±3.19	99.35±1.75	97.65±4.29	95.53±8.21	92.63±7.18	**99.37±0.81**	98.39±1.44	98.91±0.94
OA (%)	96.23±0.72	94.73±0.91	86.00±1.80	93.18±1.00	96.20±0.85	87.30±1.39	92.09±1.23	94.56±1.04	96.31±1.00	76.83±4.52	79.77±1.67	92.01±1.18	95.90±0.92	**98.28±0.39**
AA (%)	95.18±1.22	95.29±0.77	80.51±2.57	90.63±1.74	95.54±0.85	84.17±2.60	87.25±5.27	92.81±1.64	95.53±1.70	71.09±9.65	74.46±2.34	90.12±1.88	94.97±1.03	**97.90±0.51**
$\kappa \times 100$	94.99±0.96	92.97±1.23	81.31±2.44	90.95±1.33	94.94±1.14	83.07±1.86	89.44±1.66	92.77±1.38	95.10±1.33	67.84±7.26	72.61±2.33	89.36±1.59	94.56±1.23	**97.72±0.52**

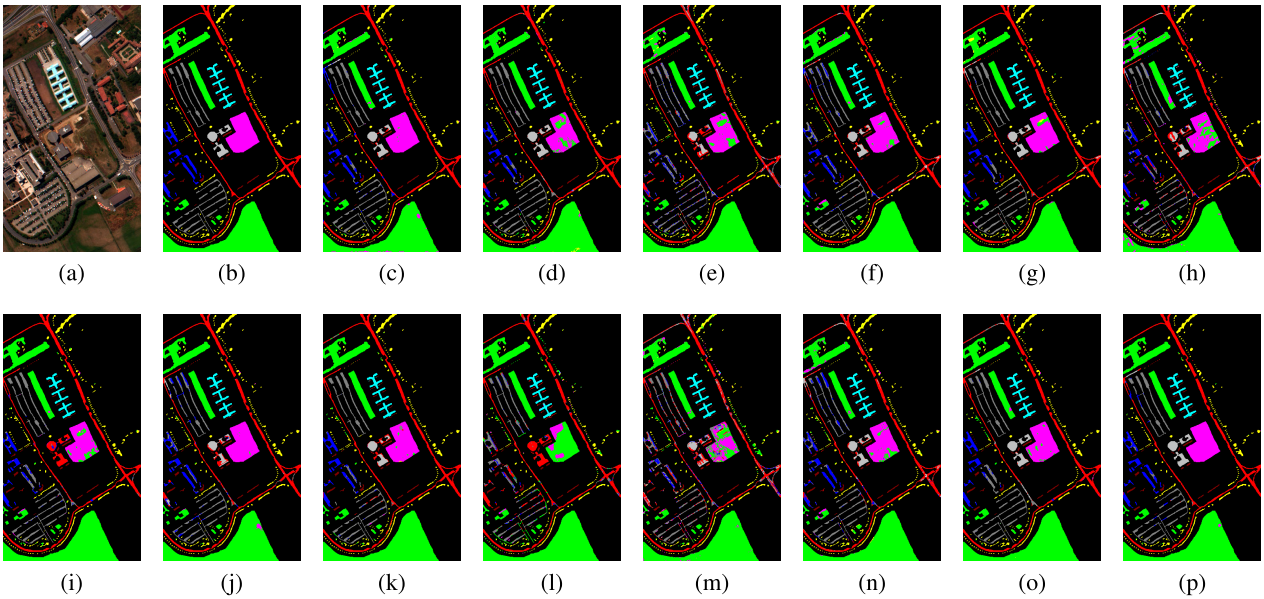


Fig. 10. Classification maps of the PU dataset with the training set ratio of 0.5%. The OA value of each method is listed in parentheses in the subfigures. (a) False color image; (b) ground truth; (c) A^2S^2K-ResNet (96.23%); (d) AMS-M2ESL (94.73%); (e) CDCNN (86.00%); (f) GMA-Net (93.18%); (g) GSC-ViT (96.20%); (h) HybridSN (87.30%); (i) LGG-CNN (92.09%); (j) MHCFormer (94.56%); (k) MS^2T^2 (96.31%); (l) PyResNet (76.83%); (m) SpectralFormer (79.77%); (n) SQSFormer (92.01%); (o) SSRN (95.90%); and (p) S^3FCN (98.28%).

(vineyard-untrained). This indicates that the proposed method can identify the trivial differences in the spectrum despite the similarity of the land objects. In addition, visualization results in Fig. 11 indicate the proposed method achieves more homogeneous labeling of large landmarks with identical features, such as C8 and C15.

3) Results of the KSC Dataset: Classification results are shown in Table V and Fig. 12. The per-class accuracy of the proposed method ranked top two in most land classes, and the overall metrics of OA, AA, and κ exceed the second place by 0.97%, 1.56%, and 1.08%, respectively. The KSC dataset also covers the visible and infrared spectra. As indicated in the result data, our method effectively employs this data augmentation through the focused analysis on the spectral domain to enhance the classification performance.

4) Results of the IP Dataset: Classification results are shown in Table VI and Fig. 13. In Table VI, our method leads the per-class accuracy in most categories, except for C7 (grass) and C9 (oats). Both classes have significantly fewer samples than other classes, making the algorithm's recognition difficult, especially when source HSI suffers from noises. Despite the data limitation, our method still gives improved classification accuracy compared to other methods and is ranked top in all overall metrics. The OA, AA, and κ from our method surpasses the second place by 1.99%, 0.70%, and 2.27%, respectively. Although the IP dataset has more complex landform arrangements compared to the SV dataset, the visualization results in Fig. 13 demonstrate that our method can still achieve good distinction of the land-cover borders, as well as much less mislabeling compared with other methods.

5) Results of the HT Dataset: Classification results are shown in Table VII and Fig. 14. In Table VII, our method achieves the highest per-class accuracy in more than half of the categories. Unlike the other datasets, the HT dataset is sparse, which adds complexity to the classification task. Despite this challenge, our method still ranks first in all overall metrics. Specifically, our method surpasses the second-best by 1.03% in OA, 1.89% in AA, and 1.11% in κ. Even though the HT dataset features the most complex landform arrangements among all five datasets, the visualization results in Fig. 14 show that our method provides better land-cover boundary

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TABLE IV
CLASSIFICATION RESULTS OF THE SV DATASET WITH THE TRAINING SET RATIO OF 0.5%

Class	A^2S^2K-ResNet [54]	AMS-M2ESL [53]	CDCNN [22]	GMA-Net [40]	GSC-ViT [52]	HybridSN [47]	LGG-CNN [55]	MHCFormer [26]	MS^2T^2 [27]	PyResNet [57]	SpectralFormer [25]	SQSFormer [48]	SSRN [45]	S^3FCN
C1	99.49±1.56	10.00±26.73	93.49±9.26	97.97±2.54	95.88±8.97	85.60±20.57	99.15±2.98	99.21±1.79	99.45±2.76	94.49±9.01	90.12±10.36	96.40±4.70	99.92±0.23	**99.99±0.01**
C2	**99.89±0.22**	66.57±9.89	92.71±8.28	99.02±1.17	98.93±1.53	85.75±12.62	98.58±2.71	98.96±1.47	99.71±0.81	96.15±2.52	89.43±9.22	97.63±3.11	99.81±0.47	99.87±0.22
C3	97.21±2.71	88.28±14.64	89.44±5.87	94.94±4.23	91.53±6.95	89.83±8.02	73.77±22.61	97.24±2.31	**98.18±0.96**	89.52±6.42	86.41±5.40	94.46±2.58	97.10±2.53	98.17±1.60
C4	97.89±1.55	96.45±2.47	94.67±5.17	96.50±3.10	94.36±4.45	94.65±4.33	96.11±2.13	97.18±1.94	**98.50±1.57**	97.34±2.01	95.68±3.71	94.74±4.87	97.76±2.05	98.47±1.31
C5	98.54±1.83	79.40±12.52	95.15±5.62	98.47±1.57	97.35±4.40	98.15±2.73	89.57±10.10	97.27±2.46	98.73±1.27	93.19±3.98	94.26±3.65	95.90±3.00	98.64±1.90	**99.24±1.02**
C6	99.87±0.28	99.58±0.91	98.83±2.07	99.77±0.52	99.55±0.54	99.33±1.06	99.43±1.89	99.60±0.71	99.58±0.62	99.35±0.81	98.74±1.04	99.66±0.72	99.82±0.32	**99.97±0.06**
C7	99.29±1.26	91.48±6.40	94.96±4.10	98.47±2.01	98.24±2.18	94.81±4.54	99.50±0.61	98.83±1.90	99.52±0.86	96.74±2.52	95.32±3.03	97.96±2.51	99.88±0.32	**99.91±0.21**
C8	91.20±3.12	65.07±7.87	82.50±4.25	87.73±3.49	89.87±4.68	80.16±3.64	85.50±6.81	84.87±4.95	91.70±2.12	76.72±4.39	76.10±4.11	85.80±4.27	90.81±2.64	**95.84±2.04**
C9	99.31±0.74	81.32±10.22	98.38±1.67	99.35±0.76	98.40±2.21	95.59±3.64	95.82±4.75	99.33±0.72	99.52±0.27	98.65±1.85	94.94±3.20	98.29±1.53	99.66±0.35	**99.70±0.20**
C10	96.85±2.47	83.23±9.17	90.93±4.33	95.17±4.96	92.66±5.63	89.81±6.10	89.16±7.18	96.62±2.44	96.83±3.03	89.10±4.50	86.40±5.14	93.33±4.25	97.42±1.39	**98.66±1.74**
C11	95.58±2.85	25.47±41.77	84.88±11.19	93.88±4.96	87.95±11.21	81.43±11.74	17.86±35.92	96.36±2.04	95.33±2.98	84.40±8.86	80.63±8.64	91.83±5.03	**96.86±1.92**	96.21±3.55
C12	98.37±1.73	79.45±13.58	91.59±7.25	96.57±4.88	96.86±5.20	95.30±6.31	92.90±8.79	98.47±1.86	**99.46±1.13**	93.57±5.55	90.70±7.16	96.73±3.20	98.06±2.53	99.38±1.01
C13	96.59±4.40	87.56±27.99	91.97±6.86	98.88±1.79	93.72±9.15	92.99±5.82	92.35±24.74	98.75±1.70	**99.72±0.68**	82.12±8.37	92.77±4.20	96.06±4.47	96.82±2.88	99.23±1.08
C14	97.01±3.50	90.83±18.35	86.90±10.04	97.63±3.46	95.99±5.37	94.51±6.07	91.77±12.44	97.61±4.09	97.60±2.14	91.08±6.05	90.79±5.96	94.48±7.15	98.08±2.51	**98.54±1.80**
C15	85.67±4.52	46.40±31.33	73.66±6.24	80.14±3.15	82.52±3.59	75.74±5.22	76.40±6.46	78.82±5.50	88.32±3.21	67.69±5.18	68.68±6.47	78.51±3.94	85.63±4.15	**92.29±2.91**
C16	99.02±1.54	94.85±18.44	94.77±4.84	99.39±1.12	98.54±3.12	91.20±7.96	99.74±0.48	99.01±1.43	99.84±0.58	96.98±4.85	94.93±3.82	98.75±2.32	99.80±0.61	**99.96±0.13**
OA (%)	95.24±0.68	75.86±3.01	88.51±1.59	93.31±0.78	92.96±1.28	87.46±2.06	89.91±2.25	92.87±1.45	95.98±0.53	87.05±1.36	85.64±2.00	91.86±1.00	95.37±0.74	**97.63±0.46**
AA (%)	96.99±0.52	74.12±5.41	90.93±1.62	95.87±0.67	94.52±1.62	90.30±2.05	87.35±4.74	96.13±0.76	97.62±0.38	90.44±1.48	89.12±1.88	94.41±1.10	97.25±0.49	**98.46±0.36**
$\kappa \times 100$	94.70±0.76	72.79±3.53	87.20±1.76	92.55±0.87	92.16±1.43	86.02±2.30	88.75±2.51	92.06±1.62	95.52±0.59	85.58±1.51	84.00±2.23	90.93±1.12	94.85±0.82	**97.36±0.51**

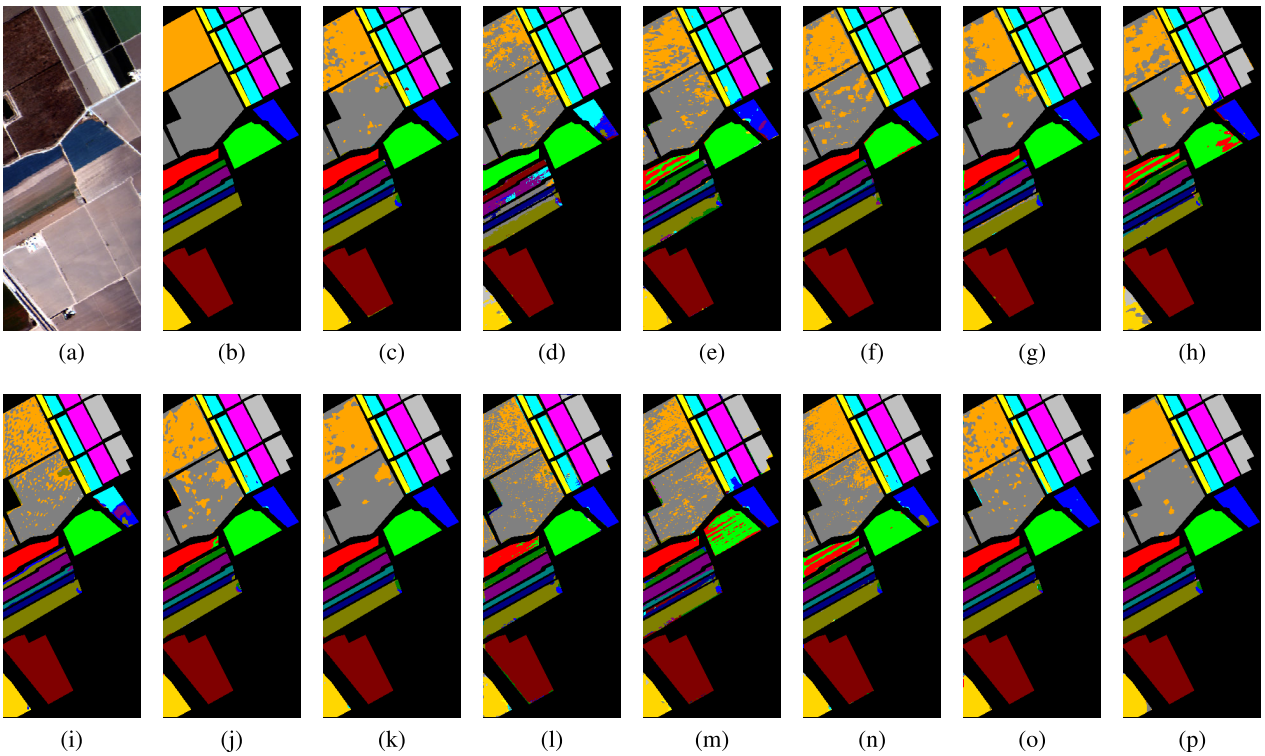


Fig. 11. Classification maps of the SV dataset with the training set ratio of 0.5%. The OA value of each method is listed in parentheses in the subfigures. (a) False color image; (b) ground truth; (c) A^2S^2K-ResNet (96.23%); (d) AMS-M2ESL (94.73%); (e) CDCNN (86.00%); (f) GMA-Net (93.18%); (g) GSC-ViT (96.20%); (h) HybridSN (87.30%); (i) LGG-CNN (92.09%); (j) MHCFormer (94.56%); (k) MS^2T^2 (96.31%); (l) PyResNet (76.83%); (m) SpectralFormer (79.77%); (n) SQSFormer (92.01%); (o) SSRN (95.90%); and (p) S^3FCN (98.28%).

distinction and significantly reduces misclassification compared to other methods.

The results demonstrate that the proposed S^3FCN can fulfill various HSIC tasks with diverse landscape distributions compared with various premier methods based on different DL frameworks. The proposed method accurately restores land-cover boundaries while maintaining classification accuracy for datasets with diverse geometric shapes, such as PU, KSC, and HT. For datasets with regularly arranged large areas, like SV and IP, the proposed method achieves high accuracy and spatial consistency in labeling large homogeneous regions. These results indicate that S^3FCN, by sequentially processing and integrating spectral and spatial domain information, can simultaneously enhance feature labeling accuracy and landform shape delineation.

B. Computation Efficiency Analysis

The computation efficiency is evaluated by considering the balance between the network's OA in classification and computation cost. The computation overhead metrics involve networks' parameter count and MACs across all five datasets and, on average, among the other 13 methods, as summarized in Table VIII. In addition, a comparison of the OA values averaged over all datasets is plotted against parameters and

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TABLE V
CLASSIFICATION RESULTS OF THE KSC DATASET WITH THE TRAINING SET RATIO OF 5%

Class	A^2S^2K-ResNet [54]	AMS-M2ESL [53]	CDCNN [22]	GMA-Net [40]	GSC-ViT [52]	HybridSN [47]	LGG-CNN [55]	MHCFormer [26]	MS^2T^2 [27]	PyResNet [57]	SpectralFormer [25]	SQSFormer [48]	SSRN [45]	S^3FCN
C1	99.25±0.95	97.11±1.48	96.52±2.49	99.56±0.94	**99.89±0.39**	97.38±2.15	98.04±1.87	98.77±1.74	99.64±0.65	97.77±1.95	97.27±1.21	98.58±1.21	99.05±0.97	99.77±0.51
C2	97.90±3.52	92.59±5.53	80.60±9.47	95.20±4.89	96.83±3.56	88.80±8.17	93.05±6.04	97.27±3.82	97.87±3.07	87.85±6.98	76.60±7.87	92.45±5.34	98.65±2.34	**98.94±2.41**
C3	88.86±6.87	85.48±8.13	65.64±10.17	82.72±7.46	95.16±5.88	84.48±6.01	81.10±5.72	87.62±5.55	95.85±3.53	68.03±9.34	59.23±6.79	82.70±8.47	94.70±5.77	**99.68±0.89**
C4	88.37±7.94	80.80±8.71	54.51±7.52	81.89±6.57	86.91±5.98	75.58±7.34	76.59±11.23	86.42±6.89	88.98±6.14	59.69±8.20	55.60±8.74	73.73±7.06	87.49±8.51	**93.74±4.42**
C5	88.08±9.23	87.06±8.86	53.53±10.20	81.09±10.52	87.06±8.63	70.42±7.14	80.97±12.40	87.80±7.07	93.08±6.96	52.49±11.00	47.03±8.33	74.66±9.65	87.68±9.84	**96.27±5.86**
C6	97.32±3.81	92.58±4.58	80.10±8.06	90.69±5.07	96.84±5.39	84.14±8.05	85.09±8.31	94.51±5.92	97.78±3.56	79.87±6.47	82.81±5.41	84.71±5.99	97.90±2.63	**99.60±0.74**
C7	96.67±4.49	89.28±8.28	76.38±14.18	86.97±10.67	94.02±8.85	79.45±10.90	83.57±18.70	92.41±7.13	95.43±7.37	80.99±10.06	81.14±10.42	83.37±12.23	95.93±7.85	**99.05±4.56**
C8	98.72±1.39	95.61±5.52	82.03±8.20	98.62±1.51	98.99±1.00	93.71±3.89	98.26±1.39	98.37±1.67	99.18±1.21	83.70±7.86	79.24±10.27	98.39±1.80	98.20±1.80	**99.49±0.71**
C9	99.19±1.62	97.26±1.56	87.80±7.11	99.81±0.43	99.78±0.38	94.54±3.47	99.75±0.71	99.83±0.33	99.65±0.36	87.79±5.35	78.68±6.48	98.88±1.16	99.86±0.25	**99.93±0.24**
C10	99.05±2.04	99.42±0.33	86.26±6.12	99.94±0.34	99.44±1.36	96.48±4.64	99.99±0.05	99.56±1.32	**100.00±0.00**	95.04±3.77	82.94±9.80	99.52±1.04	99.20±1.31	**100.00±0.00**
C11	99.34±1.22	96.32±2.72	97.33±3.45	99.29±1.20	99.04±1.44	99.25±1.47	99.21±1.55	**99.90±0.20**	98.96±1.70	99.09±1.53	99.58±0.87	98.01±2.96	99.16±1.47	99.64±1.13
C12	99.34±0.88	97.49±2.10	90.17±5.58	99.13±0.92	98.65±1.53	96.48±2.63	99.48±0.75	98.70±1.65	99.32±1.29	93.24±3.10	94.91±2.88	98.77±0.81	99.50±1.06	**99.80±0.48**
C13	99.99±0.04	99.91±0.30	99.55±0.84	**100.00±0.00**	**100.00±0.00**	99.64±0.56	99.99±0.02	**100.00±0.00**	99.94±0.30	99.76±0.38	99.42±0.85	99.74±0.31	**100.00±0.00**	**100.00±0.00**
OA (%)	97.57±0.58	95.20±1.20	86.83±1.80	96.31±0.80	97.80±0.70	93.11±1.30	95.29±0.98	97.24±0.81	98.34±0.53	88.76±1.28	85.55±2.56	94.77±0.97	97.89±0.86	**99.31±0.42**
AA (%)	96.31±0.77	93.15±1.51	80.80±2.27	93.45±1.61	96.35±1.19	89.26±1.80	91.93±2.41	95.47±1.24	97.36±1.00	83.48±1.90	79.57±3.05	91.04±1.96	96.72±1.44	**98.92±0.75**
$\kappa \times 100$	97.30±0.65	94.66±1.33	85.32±2.01	95.89±0.89	97.55±0.78	92.33±1.45	94.76±1.09	96.93±0.91	98.16±0.59	87.49±1.42	83.91±2.85	94.17±1.08	97.65±0.95	**99.24±0.46**

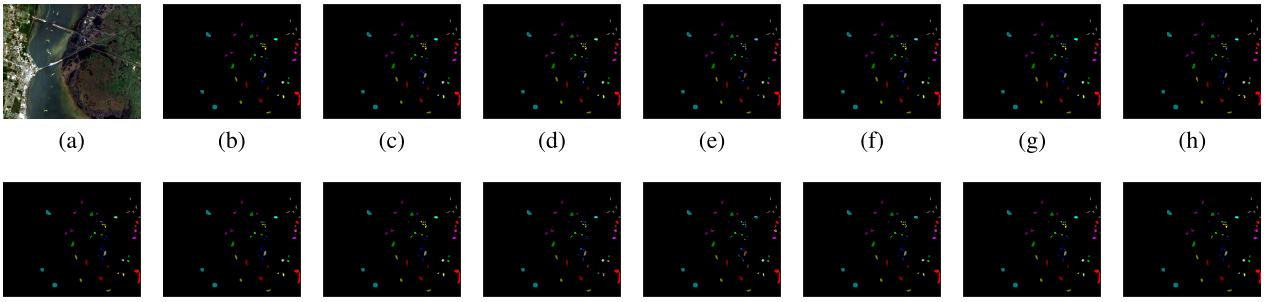


Fig. 12. Classification maps of the KSC dataset with the training set ratio of 5%. The OA value of each method is listed in parentheses in the subfigures. (a) False color image; (b) ground truth; (c) A^2S^2K-ResNet (96.23%); (d) AMS-M2ESL (94.73%); (e) CDCNN (86.00%); (f) GMA-Net (93.18%); (g) GSC-ViT (96.20%); (h) HybridSN (87.30%); (i) LGG-CNN (92.09%); (j) MHCFormer (94.56%); (k) MS^2T^2 (96.31%); (l) PyResNet (76.83%); (m) SpectralFormer (79.77%); (n) SQSFormer (92.01%); (o) SSRN (95.90%); and (p) S^3FCN (98.28%).

TABLE VI
CLASSIFICATION RESULTS OF THE IP DATASET WITH THE TRAINING SET RATIO OF 3%

Class	A^2S^2K-ResNet [54]	AMS-M2ESL [53]	CDCNN [22]	GMA-Net [40]	GSC-ViT [52]	HybridSN [47]	LGG-CNN [55]	MHCFormer [26]	MS^2T^2 [27]	PyResNet [57]	SpectralFormer [25]	SQSFormer [48]	SSRN [45]	S^3FCN
C1	91.69±10.30	73.65±41.06	50.54±19.17	89.95±11.98	89.01±13.42	75.38±19.47	0±0	**96.52±7.04**	95.21±6.27	65.88±23.40	74.08±17.50	76.63±15.43	82.37±18.06	95.31±7.21
C2	91.44±4.38	87.47±4.03	60.71±4.73	86.42±4.66	89.10±5.71	72.88±5.79	80.14±6.31	90.69±3.37	92.54±2.90	64.38±5.15	57.08±4.53	82.84±4.78	91.18±4.09	**96.41±1.92**
C3	92.17±4.20	91.33±5.20	48.20±6.14	88.50±4.88	91.00±4.52	62.31±8.69	83.54±6.79	90.73±4.01	94.47±3.18	50.50±4.87	45.22±6.28	84.71±4.78	90.71±5.37	**96.26±1.85**
C4	92.25±8.05	84.40±7.62	41.08±12.17	89.82±6.04	89.19±8.41	63.89±15.56	64.48±22.20	95.46±3.84	93.59±5.61	55.40±17.67	59.70±10.78	79.24±8.81	88.34±8.25	**95.62±4.91**
C5	97.82±2.70	**98.72±2.19**	86.92±6.82	93.87±4.59	96.48±3.55	87.72±6.92	90.00±5.04	98.10±2.50	98.20±1.69	75.07±12.70	75.66±7.80	95.73±2.92	98.01±2.87	98.69±1.64
C6	97.08±2.45	**98.57±1.55**	84.46±6.50	96.44±2.78	96.07±2.65	88.90±5.17	94.65±3.21	97.46±2.07	96.57±2.38	69.00±8.28	66.94±5.49	95.67±3.31	96.58±2.37	98.30±1.72
C7	84.56±15.05	**93.06±11.49**	39.65±20.20	66.95±16.35	59.74±19.52	68.15±20.09	2.00±10.77	79.69±20.74	67.32±20.83	39.45±22.25	67.30±14.81	73.05±18.62	86.10±19.88	80.32±21.51
C8	98.67±1.53	96.79±3.57	91.36±3.85	97.66±1.69	98.29±2.06	94.46±3.25	93.49±1.82	99.18±1.42	99.71±0.62	93.59±4.04	90.57±2.64	97.13±1.65	98.28±1.97	**99.78±0.81**
C9	76.21±16.23	82.76±14.68	36.75±17.07	69.55±18.00	69.56±16.57	39.57±16.78	6.00±19.01	**94.54±7.61**	76.11±11.56	34.30±18.32	48.42±18.15	72.70±17.32	73.07±17.50	75.09±25.85
C10	90.86±5.06	92.20±3.65	62.74±9.05	85.47±4.97	87.73±5.81	75.85±4.79	83.63±5.98	89.10±5.36	92.07±3.47	65.24±9.62	56.07±6.44	85.15±5.27	89.41±4.40	**93.38±3.72**
C11	91.61±3.02	90.58±2.19	71.32±5.23	89.05±3.26	93.68±2.46	81.14±4.66	86.12±4.03	92.66±2.46	95.11±1.95	69.14±4.58	67.62±3.72	87.31±3.63	92.53±2.63	**97.59±0.93**
C12	89.33±7.04	84.83±6.81	47.85±9.01	81.15±7.42	85.23±7.50	56.15±9.32	71.56±8.84	88.77±4.29	94.49±2.45	49.27±7.94	46.88±6.03	77.16±7.71	89.19±6.62	**94.64±4.40**
C13	98.16±2.31	**99.90±0.56**	74.95±11.66	95.22±4.82	97.00±3.44	85.74±11.70	93.94±7.54	98.29±2.75	98.82±1.63	77.47±13.82	75.32±8.85	97.53±1.52	98.07±2.19	98.29±2.61
C14	97.03±1.67	93.72±2.57	89.15±4.66	95.13±3.47	95.56±3.02	92.65±2.91	95.12±2.15	96.94±2.60	97.32±1.78	89.12±4.46	85.45±4.29	93.63±2.20	97.03±1.91	**98.16±0.94**
C15	93.47±4.53	90.73±5.23	67.13±11.67	89.27±4.24	88.45±5.76	81.07±9.31	83.69±6.99	94.51±3.04	92.51±4.17	62.47±11.16	59.76±9.01	86.97±6.18	91.24±5.94	**96.05±2.77**
C16	91.85±5.26	91.48±4.85	74.31±14.44	89.07±7.89	89.30±8.99	88.51±9.26	53.69±47.33	**94.52±5.02**	90.97±8.31	87.30±12.08	86.08±13.75	94.50±3.98	92.55±7.98	94.46±5.42
OA (%)	93.03±1.03	91.52±0.88	69.55±2.01	89.51±1.43	91.66±1.56	78.64±2.29	85.53±1.14	93.14±1.14	94.73±0.91	68.11±1.42	65.69±2.36	87.64±1.51	92.64±1.18	**96.72±0.66**
AA (%)	92.14±1.79	90.64±3.34	64.20±3.53	87.72±2.35	88.46±2.46	75.90±3.24	67.63±4.95	93.57±1.70	92.19±1.77	65.47±2.75	66.38±3.50	86.25±2.34	90.92±2.21	**94.27±2.73**
$\kappa \times 100$	92.04±1.18	90.31±1.00	65.09±2.31	88.01±1.64	90.49±1.77	75.59±2.62	83.47±1.31	92.16±1.30	93.99±1.04	63.41±1.68	60.50±2.76	85.88±1.74	91.60±1.34	**96.26±0.75**

MACs in a bubble graph, as shown in Fig. 15, with all methods' averaged OA values listed above. The bubble size exhibits the original OA values projected into a rescaled exponential space for a more evident illustration. Our proposed network's averaged OA of 97.46% surpasses the second-highest result from MS^2T^2 by 1.52%. The comparison indicates our method can achieve the highest accuracy while maintaining relative lightweight in terms of restrained parameters and MACs.

C. Ablation Study

1) Contribution of Attention Modules: We performed ablation studies to examine the contribution of the attention module, S^3AM, added within the second stage of the main frame for the spectral-weighed spatial feature extraction to verify its effectiveness. As illustrated in Fig. 3, the attention module is applied in the spatial feature extraction subnetwork through either a long-jumping connection between input and output or short connections between convolution layers. Thus, we considered four cases: using only the CNN backbone with no attention, using attention at the input-output connection (S^3AM-global), at interlayer connections (S^3AM-layer), and both pathways (S^3AM-all). Table IX summarizes the results across all datasets. Embedding S^3AM in both pathways achieved the highest OA score, exceeding the no-attention case

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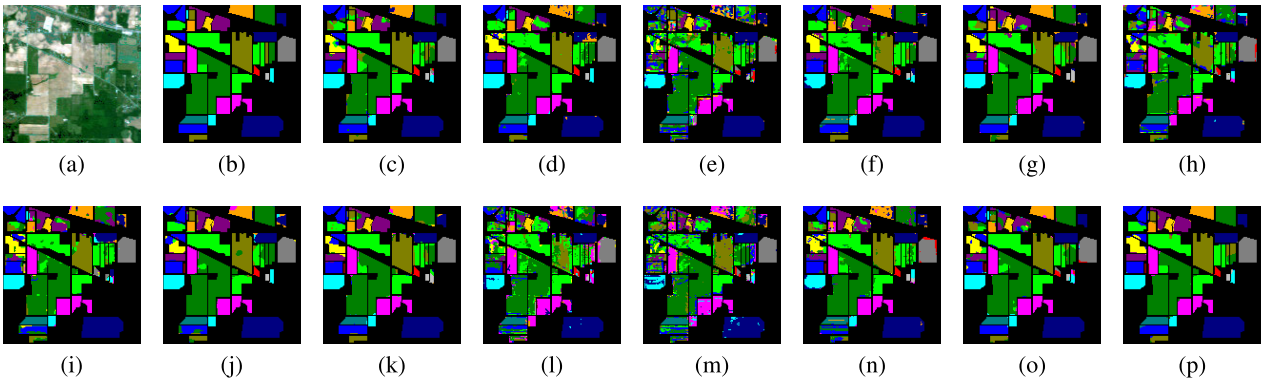


Fig. 13. Classification maps of the IP dataset with the training set ratio of 3%. The OA value of each method is listed in parentheses in the subfigures. (a) False color image; (b) ground truth; (c) A^2S^2K-ResNet (96.23%); (d) AMS-M2ESL (94.73%); (e) CDCNN (86.00%); (f) GMA-Net (93.18%); (g) GSC-ViT (96.20%); (h) HybridSN (87.30%); (i) LGG-CNN (92.09%); (j) MHCFormer (94.56%); (k) MS^2T^2 (96.31%); (l) PyResNet (76.83%); (m) SpectralFormer (79.77%); (n) SQSFormer (92.01%); (o) SSRN (95.90%); and (p) S^3FCN (98.28%).

TABLE VII

CLASSIFICATION RESULTS OF THE HT DATASET WITH THE TRAINING SET RATIO OF 2%

Class	A^2S^2K-ResNet [54]	AMS-M2ESL [53]	CDCNN [22]	GMA-Net [40]	GSC-ViT [52]	HybridSN [47]	LGG-CNN [55]	MHCFormer [26]	MS^2T^2 [27]	PyResNet [57]	SpectralFormer [25]	SQSFormer [48]	SSRN [45]	S^3FCN
C1	95.39±2.77	93.58±3.90	91.20±7.08	89.47±6.75	92.41±4.95	88.99±6.37	93.14±6.21	93.61±4.90	96.57±3.19	89.96±6.45	90.47±5.41	91.26±3.97	95.73±3.54	**97.67±2.29**
C2	96.83±3.05	95.84±3.38	85.06±6.65	93.66±5.93	93.51±5.36	90.39±6.14	89.15±8.82	95.00±4.52	**97.46±3.17**	84.87±7.79	86.86±5.20	94.45±4.45	96.72±3.66	97.32±2.89
C3	99.51±1.30	99.78±0.65	80.20±12.12	98.86±2.16	98.48±2.86	94.15±5.49	97.58±5.47	97.93±3.95	99.95±0.20	89.62±7.25	87.79±5.20	98.35±1.76	**99.96±0.12**	99.84±0.42
C4	97.86±2.35	96.46±4.13	93.37±4.15	96.92±2.98	97.66±3.33	93.53±3.42	97.44±3.02	97.57±3.23	94.93±3.27	94.28±5.38	91.01±4.13	97.67±3.23	98.41±2.56	**98.49±1.34**
C5	94.88±3.76	97.28±2.56	93.40±4.61	96.91±3.10	**97.39±2.97**	95.28±2.55	94.59±3.11	95.26±3.49	97.37±2.90	92.69±3.97	86.85±4.65	96.22±3.17	96.86±2.45	97.36±2.59
C6	98.25±3.64	97.87±3.43	77.13±17.75	95.93±5.15	96.45±5.01	87.78±8.85	97.36±6.13	99.16±0.96	98.05±3.44	88.05±10.88	76.11±12.51	97.13±2.72	99.43±1.72	**99.57±0.91**
C7	90.76±3.76	86.68±3.91	79.01±6.29	88.97±4.43	88.78±5.39	84.15±3.43	90.04±4.31	89.95±3.36	90.93±4.09	81.86±6.17	80.53±4.21	88.55±3.22	91.99±3.67	**93.09±3.80**
C8	93.96±3.72	90.99±3.98	91.37±6.41	93.56±5.24	92.44±9.04	87.32±5.30	93.81±6.57	94.28±2.88	96.62±3.87	76.39±6.65	73.82±6.29	87.68±4.35	94.01±3.61	**96.80±2.86**
C9	89.70±3.31	85.56±5.16	76.40±7.00	87.17±4.92	85.96±4.99	86.04±4.86	87.70±5.37	88.69±3.96	91.62±4.30	76.90±5.45	74.37±4.77	85.49±4.42	89.36±3.99	**92.14±3.97**
C10	86.77±4.85	86.37±4.22	70.98±8.23	80.02±5.83	85.27±5.96	78.65±5.58	77.74±6.52	85.55±4.70	88.55±4.68	68.79±6.23	67.43±5.20	80.66±4.56	87.39±3.64	**91.37±3.41**
C11	92.68±4.58	92.68±2.02	74.48±6.84	90.52±5.14	92.43±2.95	78.87±6.76	90.16±2.88	91.80±4.27	95.01±3.27	71.69±8.11	71.75±6.13	85.96±5.53	92.70±4.49	**95.10±3.31**
C12	87.62±3.96	83.23±5.34	76.87±6.53	83.23±4.81	85.21±4.33	81.64±5.63	81.96±4.18	85.25±5.33	**92.76±3.57**	71.03±5.40	65.85±5.84	81.88±3.95	89.84±4.59	92.32±3.60
C13	94.24±3.85	91.38±7.10	87.17±8.47	92.82±3.48	91.23±6.92	88.08±5.75	**94.61±3.59**	90.27±6.31	90.71±4.03	82.45±13.17	78.84±9.42	93.28±4.22	94.42±4.71	91.98±4.74
C14	96.98±5.08	97.12±2.12	82.63±10.10	99.79±0.67	96.02±6.44	85.85±8.26	98.39±2.17	99.14±1.64	**100.00±0.00**	78.62±10.15	80.15±7.43	99.34±1.16	97.66±3.11	98.03±3.19
C15	96.62±2.39	96.69±3.92	90.54±4.98	94.89±3.00	96.12±2.37	95.54±3.46	95.42±2.65	96.84±1.74	93.86±3.88	92.88±4.01	86.54±7.00	96.08±2.13	**97.26±1.69**	96.73±2.66
OA (%)	93.19±0.94	91.60±0.96	82.24±2.21	90.62±1.01	91.40±1.22	86.84±1.62	90.02±1.22	92.21±1.38	94.35±1.06	80.72±1.59	78.97±1.89	90.01±0.99	93.86±0.91	**95.38±0.77**
AA (%)	94.14±0.77	92.77±0.95	83.32±2.64	92.18±0.98	92.62±0.87	87.75±1.66	91.94±1.10	93.35±1.17	94.96±0.80	82.67±1.99	79.89±2.29	91.60±0.84	94.78±0.82	**95.85±0.63**
$\kappa \times 100$	92.64±1.01	90.91±1.03	80.80±2.39	89.86±1.09	90.71±1.32	85.77±1.76	89.21±1.32	91.58±1.49	93.89±1.15	79.12±1.73	77.22±2.05	89.19±1.07	93.36±0.98	**95.00±0.83**

TABLE VIII

COMPARISON AMONG DIFFERENT NETWORKS ON NUMBERS OF PARAMETERS (PARAMS) AND MACS OVER EACH DATASET AND ON AVERAGE OF ALL DATASETS, WITH THE LOWEST VALUES IN BOLD

Method	PU		SV		KSC		IP		HT		Averaged	
	Params (K)	MACs (M)	Params (K)	MACs (M)	Params (K)	MACs (M)	Param (K)	MACs (M)	Params (K)	MACs (M)	Params (K)	MACs (M)
A^2S^2K-ResNet	215.65	85.84	368.26	170.99	325.52	147.15	362.16	167.58	276.80	119.90	309.68	138.29
AMS-M2ESL	10731.80	11.36	41896.21	43.54	31202.46	32.53	40272.11	41.87	20911.08	21.91	29002.73	30.24
CDCNN	703.63	56.78	1216.39	99.29	1063.51	87.50	1198.89	97.61	943.76	74.07	1025.24	83.05
GMA-Net	905.38	25.29	949.83	**5.25**	986.64	32.01	1013.83	34.25	950.76	29.03	961.29	25.17
GSC-ViT	149.13	6.11	174.83	8.20	167.64	**7.62**	623.52	27.79	159.76	6.96	254.98	11.34
HybridSN	2691.24	261.93	4510.13	517.95	4005.75	446.97	4438.13	507.81	3430.00	365.86	3815.05	420.10
LGG-CNN	18314.52	83.12	18457.93	92.55	18413.67	89.93	18453.43	92.18	18386.18	86.95	18405.15	88.95
MHCFormer	1996.69	161.57	3814.92	316.03	3310.82	273.21	3742.92	309.91	2734.89	224.27	3120.05	257.00
MS^2T^2	130.48	9.67	475.75	37.03	358.85	27.73	457.84	35.60	245.16	18.65	333.62	25.74
PyResNet	21556.59	34.94	21876.42	42.87	21786.16	40.67	21864.17	42.55	21691.17	38.16	21754.90	39.84
SpectralFormer	173.10	**4.28**	363.02	16.28	298.33	12.17	353.24	15.66	235.98	8.21	284.74	11.32
SQSFormer	213.41	4.95	271.55	9.67	255.23	8.36	269.30	**9.48**	237.61	**6.87**	249.42	**7.87**
SSRN	211.46	79.74	361.63	158.66	319.56	136.56	355.63	155.51	271.61	111.31	303.98	128.36
S^3FCN	**92.56**	40.68	**103.15**	76.81	**100.06**	66.69	**102.76**	75.36	**97.17**	55.13	**99.14**	62.93

by 0.93% on average. This demonstrates that incorporating spectral features enhances spatial feature extraction by maintaining spectral correlation, thereby improving classification accuracy. Adding S^3AM either globally or between layers also improved results, with the best performance observed when used in both pathways.

2) Influence of Attention Strategies: Table X presents the experimental results evaluating the sequential integration of spectral and spatial attention in S^3AM. The results indicate that using both attention blocks in a consecutive spectral–spatial manner outperforms using either block individually, with an OA score improvement of 0.75% over using

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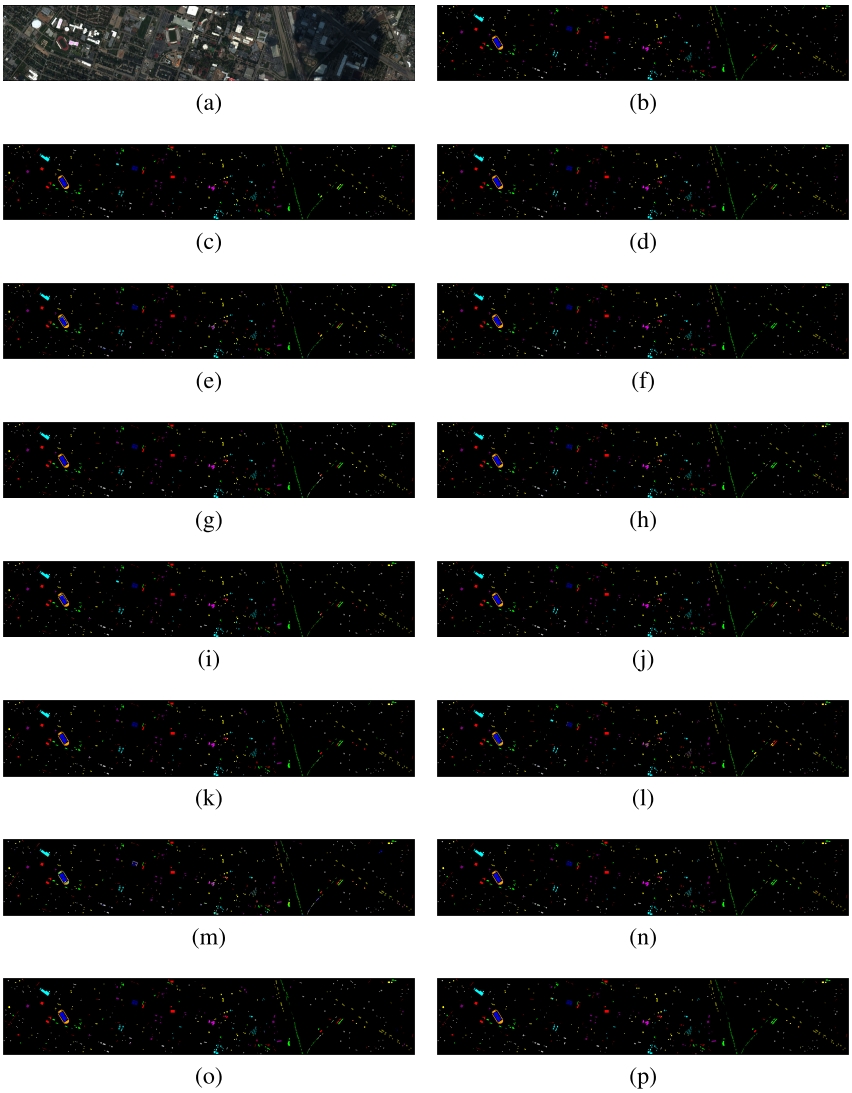


Fig. 14. Classification maps of the HT dataset with the training set ratio of 2%. The OA value of each method is listed in parentheses in the subfigures. (a) False color image; (b) ground truth; (c) A^2S^2K-ResNet (96.23%); (d) AMS-M2ESL (94.73%); (e) CDCNN (86.00%); (f) GMA-Net (93.18%); (g) GSC-ViT (96.20%); (h) HybridSN (87.30%); (i) LGG-CNN (92.09%); (j) MHCFormer (94.56%); (k) MS^2T^2 (96.31%); (l) PyResNet (76.83%); (m) SpectralFormer (79.77%); (n) SQSFormer (92.01%); (o) SSRN (95.90%); and (p) S^3FCN (98.28%).

TABLE IX

COMPARISON OF OA (IN %) FOR DIFFERENT S^3AM CONNECTIONS IN THE MAIN NETWORK, WITH THE HIGHEST VALUES IN BOLD

Attention Connection	PU	SV	KSC	IP	HT	Average
Only Backbone	97.24	96.88	98.45	95.27	93.98	96.36
Backbone+S^3AM-global	97.59	97.18	98.81	95.84	94.60	96.80
Backbone+S^3AM-layer	97.90	97.32	99.04	96.31	94.99	97.11
Backbone+S^3AM-all	**98.28**	**97.63**	**99.31**	**96.72**	**95.38**	**97.46**

TABLE X

COMPARISON OF OA (IN %) USING DIFFERENT ATTENTION STRATEGY IN S^3AM, WITH THE HIGHEST VALUES IN BOLD

Attention Strategy	PU	SV	KSC	IP	HT	Average
No Attention	97.24	96.88	98.45	95.27	93.98	96.36
Only Spectral	97.67	97.11	98.77	95.63	94.38	96.71
Only Spatial	97.78	97.25	98.91	95.91	94.71	96.91
Integrated	**98.28**	**97.63**	**99.31**	**96.72**	**95.38**	**97.46**

only the spectral block and 0.55% over using only the spatial block.

3) Influence of Training Patch Sizes: The size of the training patches can affect the convolution receptive field and affect the network's classification performance. To evaluate this effect, HSI patches with spatial dimensions of 5×5, 7×7, 9×9, 11×11, and 13×13 were tested. Results shown in Fig. 16 guide patch size selection. Overall classification accuracy generally improves with larger patches, but convergence occurs midway. Considering stability across datasets and the tradeoff between performance, parameters, and MACs, the optimal patch size is selected as 9×9.

4) Influence of Training Set Ratios: The training set ratio, representing the percentage of samples used for training, is another crucial factor affecting model performance. We conducted comprehensive tests with the results summarized in

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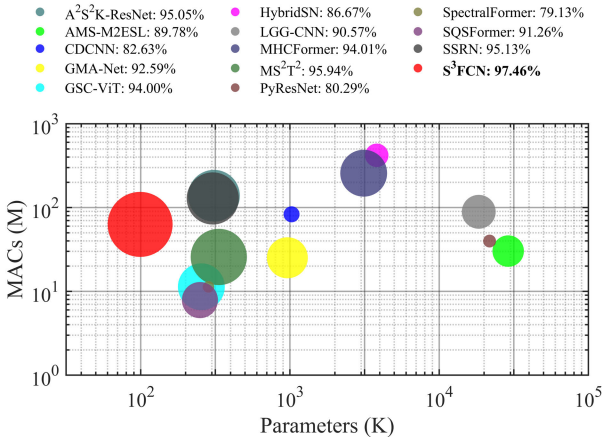


Fig. 15. Comparison of different networks considering the classification accuracy against parameters and MACs. The bubble size represents the OA value that averaged over all datasets, which are listed above the graph. The proposed method is marked with bold text.

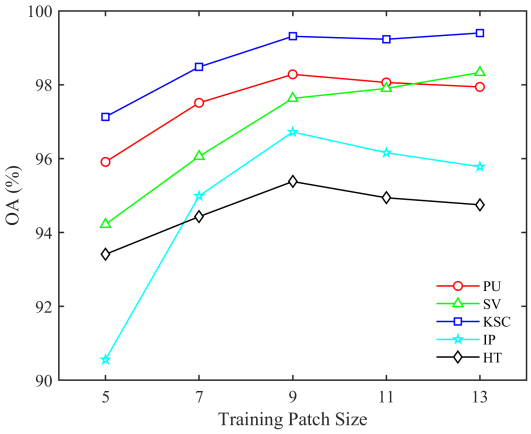


Fig. 16. Comparison of the training patch size's influence over the network's classification accuracy on each dataset.

Fig. 17, which were collected on all datasets for training set ratios of {0.5%, 1%, 3%, 5%, 10%, 15%}. Across datasets, all methods show an increasing trend in OA values with larger training ratios, and the proposed S^{3}FCN consistently leads. Performance converges at higher ratios, which is expected as the amount of training samples increases. To balance classification performance and generalization, our method selects a training set ratio within the early convergence range of 0.5%–5%, based on evaluations across different datasets.

VI. Conclusion

This article presents an HSIC algorithm, S^{3}FCN, which performs multidomain deep feature extraction using a sequentially connected and cross-weighted spectral–spatial feature convolution network. The network utilizes 1D- and 2D-DSC kernels as the basic computing units for spectral and spatial domains, respectively, offering greater efficiency than conventional convolution. The proposed architecture consists of two successive subnetworks that extract spectral and spatial features consecutively. This effectively reduces cross-domain interference while generating integrated feature maps to

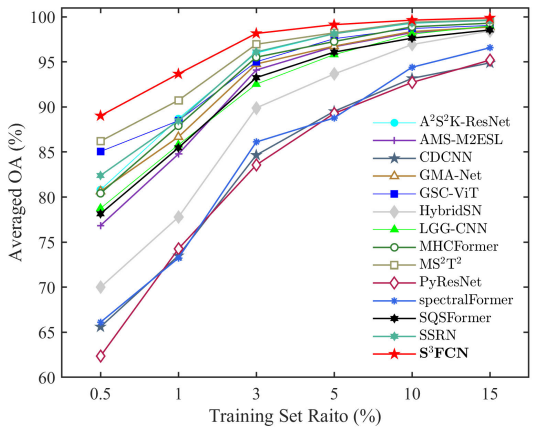


Fig. 17. Comparison of the training set ratio's influence over the network's classification accuracy. The OA value is averaged across all datasets. The proposed method is marked with bold text.

enhance classification. Furthermore, the S^{3}AM attention module computes spectral and spatial attention in sequence to achieve integration of cross-domain attention during spatial feature extraction while maintaining computational efficiency. The proposed method was evaluated on five benchmark datasets and compared with 13 SOTA HSIC methods based on CNN and transformer backbones. The results demonstrate that S^{3}FCN consistently achieves high per-class and overall classification accuracy across all datasets. Statistical evaluations of parameter count and MACs show that the proposed network achieves a well-balanced tradeoff between accuracy and computational cost. Ablation studies further verify the effectiveness of the attention mechanism and identify optimal training conditions.

This study highlights the advantages of building a CNN backbone that separates and sequentially integrates spectral and spatial feature extraction. To further optimize the framework, low-rank techniques like matrix decomposition and tensor factorization can be applied to reduce attention-weight matrix complexity, mitigating the computational burden when using deeper layers. Furthermore, imbalanced sample distribution in the spatial domain remains a practical challenge, and our future research will consider elaborated studies of the multiscale decomposition of the spatial features to explore improved integration of local–global spatial attention to enhance the network's robustness and generalizability when dealing with long-range distributed and unbalanced data.

References

[1] D. Haboudane, "Hyperspectral vegetation indices and novel algorithms for predicting green LAI of crop canopies: Modeling and validation in the context of precision agriculture," *Remote Sens. Environ.*, vol. 90, no. 3, pp. 337–352, Apr. 2004.

[2] C. A. Bishop, J. G. Liu, and P. J. Mason, "Hyperspectral remote sensing for mineral exploration in Pulang, Yunnan Province, China," *Int. J. Remote Sens.*, vol. 32, no. 9, pp. 2409–2426, May 2011.

[3] T. Sankey, J. Donager, J. McVay, and J. B. Sankey, "UAV LiDAR and hyperspectral fusion for forest monitoring in the southwestern USA," *Remote Sens. Environ.*, vol. 195, pp. 30–43, Jun. 2017.

[4] S. Wang et al., "Using soil library hyperspectral reflectance and machine learning to predict soil organic carbon: Assessing potential of airborne and spaceborne optical soil sensing," *Remote Sens. Environ.*, vol. 271, Mar. 2022, Art. no. 112914.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 10:49:03 UTC from IEEE Xplore. Restrictions apply.

[5] M. Ahmad et al., "Hyperspectral image classification—Traditional to deep models: A survey for future prospects," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 968–999, 2022.

[6] M. Sugiyama, "Dimensionality reduction of multimodal labeled data by local Fisher discriminant analysis," *J. Mach. Learn. Res.*, vol. 8, no. 37, pp. 1027–1061, May 2007.

[7] T. V. Bandos, L. Bruzzone, and G. Camps-Valls, "Classification of hyperspectral images with regularized linear discriminant analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 3, pp. 862–873, Mar. 2009.

[8] A. Villa, J. A. Benediktsson, J. Chanussot, and C. Jutten, "Hyperspectral image classification with independent component discriminant analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 12, pp. 4865–4876, Dec. 2011.

[9] G. Licciardi, P. R. Marpu, J. Chanussot, and J. A. Benediktsson, "Linear versus nonlinear PCA for the classification of hyperspectral data based on the extended morphological profiles," *IEEE Geosci. Remote Sens. Lett.*, vol. 9, no. 3, pp. 447–451, May 2012.

[10] G. Taskin, H. Kaya, and L. Bruzzone, "Feature selection based on high dimensional model representation for hyperspectral images," *IEEE Trans. Image Process.*, vol. 26, no. 6, pp. 2918–2928, Jun. 2017.

[11] E. K. Nakao and A. L. M. Levada, "Unsupervised learning and feature extraction in hyperspectral imagery," in *Artificial Intelligence and Lecture Notes in Bioinformatics*. Cham, Switzerland: Springer, 2020, pp. 792–806.

[12] A. Plaza et al., "Recent advances in techniques for hyperspectral image processing," *Remote Sens. Environ.*, vol. 113, pp. S110–S122, Apr. 2009.

[13] Q. Zou, L. Ni, T. Zhang, and Q. Wang, "Deep learning based feature selection for remote sensing scene classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 11, pp. 2321–2325, Nov. 2015.

[14] Z. Qiu, J. Chen, Y. Zhao, S. Zhu, Y. He, and C. Zhang, "Variety identification of single Rice seed using hyperspectral imaging combined with convolutional neural network," *Appl. Sci.*, vol. 8, no. 2, p. 212, Jan. 2018.

[15] H. Fırat, M. E. Asker, M. I. Bayindir, and D. Hanbay, "Spatial-spectral classification of hyperspectral remote sensing images using 3D CNN based LeNet-5 architecture," *Infr. Phys. Technol.*, vol. 127, Dec. 2022, Art. no. 104470.

[16] S. Yu, S. Jia, and C. Xu, "Convolutional neural networks for hyperspectral image classification," *Neurocomputing*, vol. 219, pp. 88–98, Jan. 2017.

[17] H. Gao, Y. Yang, C. Li, H. Zhou, and X. Qu, "Joint alternate small convolution and feature reuse for hyperspectral image classification," *ISPRS Int. J. Geo-Inf.*, vol. 7, no. 9, p. 349, Aug. 2018.

[18] J. Yue, W. Zhao, S. Mao, and H. Liu, "Spectral–spatial classification of hyperspectral images using deep convolutional neural networks," *Remote Sens. Lett.*, vol. 6, no. 6, pp. 468–477, Jun. 2015.

[19] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.

[20] J. M. Haut, M. E. Paoletti, J. Plaza, J. Li, and A. Plaza, "Active learning with convolutional neural networks for hyperspectral image classification using a new Bayesian approach," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 11, pp. 6440–6461, Nov. 2018.

[21] H. Liang and Q. Li, "Hyperspectral imagery classification using sparse representations of convolutional neural network features," *Remote Sens.*, vol. 8, no. 2, p. 99, Jan. 2016.

[22] H. Lee and H. Kwon, "Going deeper with contextual CNN for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 26, no. 10, pp. 4843–4855, Oct. 2017.

[23] E. Shelhamer, J. Long, and T. Darrell, "Fully convolutional networks for semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 4, pp. 640–651, Apr. 2017.

[24] G. Cheng, Z. Li, J. Han, X. Yao, and L. Guo, "Exploring hierarchical convolutional features for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 11, pp. 6712–6722, Nov. 2018.

[25] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615.

[26] H. Shi, Y. Zhang, G. Cao, and D. Yang, "MHCFormer: Multiscale hierarchical conv-aided fourierformer for hyperspectral image classification," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–15, 2024.

[27] Z. Meng, T. Zhang, F. Zhao, G. Chen, and M. Liang, "Multiscale super token transformer for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[28] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[29] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun. 2014.

[30] Y. Chen, X. Zhao, and X. Jia, "Spectral–spatial classification of hyperspectral data based on deep belief network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2381–2392, Jun. 2015.

[31] X. Ma, H. Wang, and J. Geng, "Spectral–spatial classification of hyperspectral image based on deep auto-encoder," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 9, pp. 4073–4085, Sep. 2016.

[32] X. Sun, F. Zhou, J. Dong, F. Gao, Q. Mu, and X. Wang, "Encoding spectral and spatial context information for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 12, pp. 2250–2254, Dec. 2017.

[33] S. Mei, J. Ji, J. Hou, X. Li, and Q. Du, "Learning sensor-specific spatial–spectral features of hyperspectral images via convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 8, pp. 4520–4533, Aug. 2017.

[34] E. Aptoula, M. C. Ozdemir, and B. Yanikoglu, "Deep learning with attribute profiles for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 12, pp. 1970–1974, Dec. 2016.

[35] B. Pan, Z. Shi, and X. Xu, "R-VCANet: A new deep-learning-based hyperspectral image classification method," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 5, pp. 1975–1986, May 2017.

[36] X. Kang, C. Li, S. Li, and H. Lin, "Classification of hyperspectral images by Gabor filtering based deep network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 4, pp. 1166–1178, Apr. 2018.

[37] P. Ghamisi, J. Plaza, Y. Chen, J. Li, and A. J. Plaza, "Advanced spectral classifiers for hyperspectral images: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 1, pp. 8–32, Mar. 2017.

[38] J. Yang, Y.-Q. Zhao, and J. C. Chan, "Learning and transferring deep joint spectral–spatial features for hyperspectral classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 8, pp. 4729–4742, Aug. 2017.

[39] S. Hao, W. Wang, Y. Ye, T. Nie, and L. Bruzzone, "Two-stream deep architecture for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 2349–2361, Apr. 2018.

[40] T. Lu, M. Liu, W. Fu, and X. Kang, "Grouped multi-attention network for hyperspectral image spectral-spatial classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5507912.

[41] C. Tao, H. Pan, Y. Li, and Z. Zou, "Unsupervised spectral–spatial feature learning with stacked sparse autoencoder for hyperspectral imagery classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 12, pp. 2438–2442, Dec. 2015.

[42] B. Pan, Z. Shi, N. Zhang, and S. Xie, "Hyperspectral image classification based on nonlinear spectral–spatial network," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 12, pp. 1782–1786, Sep. 2016.

[43] Y. He, B. Tu, B. Liu, Y. Chen, J. Li, and A. Plaza, "Hybrid multiscale spatial–spectral transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[44] C. Shi, S. Yue, and L. Wang, "Attention head interactive dual attention transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5523720.

[45] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2017.

[46] X. Yang, Y. Ye, X. Li, R. Y. K. Lau, X. Zhang, and X. Huang, "Hyperspectral image classification with deep learning models," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5408–5423, Sep. 2018.

[47] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D–2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2019.

[48] N. Chen, L. Fang, Y. Xia, S. Xia, H. Liu, and J. Yue, "Spectral query spatial: Revisiting the role of center pixel in transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5402714.

[49] F. Xu, S. Mei, G. Zhang, N. Wang, and Q. Du, "Bridging CNN and transformer with cross-attention fusion network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5522214.

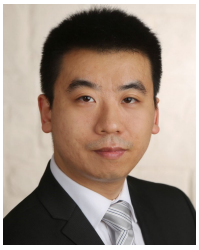
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[50] Z. Meng, L. Jiao, M. Liang, and F. Zhao, "A lightweight spectral–spatial convolution module for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[51] C. Li, B. Rasti, X. Tang, P. Duan, J. Li, and Y. Peng, "Channel-layer-oriented lightweight Spectral–Spatial network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5504214.
[52] Z. Zhao, X. Xu, S. Li, and A. Plaza, "Hyperspectral image classification using groupwise separable convolutional vision transformer network," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5511817.
[53] M. Li, W. Li, Y. Liu, Y. Huang, and G. Yang, "Adaptive mask sampling and manifold to Euclidean subspace learning with distance covariance representation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5508518.
[54] S. K. Roy, S. Manna, T. Song, and L. Bruzzone, "Attention-based adaptive spectral–spatial kernel ResNet for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7831–7843, Sep. 2021.
[55] W. Fu, K. Ding, X. Kang, and D. Wang, "Local–global gated convolutional neural network for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.
[56] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[57] M. E. Paoletti, J. M. Haut, R. Fernandez-Beltran, J. Plaza, A. J. Plaza, and F. Pla, "Deep pyramidal residual networks for spectral–spatial hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 740–754, Feb. 2019.
[58] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2261–2269.
[59] W. Ma, Q. Yang, Y. Wu, W. Zhao, and X. Zhang, "Double-branch multi-attention mechanism network for hyperspectral image classification," *Remote Sens.*, vol. 11, no. 11, p. 1307, Jun. 2019.
[60] R. Li, S. Zheng, C. Duan, Y. Yang, and X. Wang, "Classification of hyperspectral image based on double-branch dual-attention mechanism network," *Remote Sens.*, vol. 12, no. 3, p. 582, Feb. 2020.
[61] F. Chollet, "Xception: Deep learning with depthwise separable convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1800–1807.



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